A local hyperbolic shift with exact scri nullness

Spherical prototype, boundary closure, and qualification failures

Numerical formulation note

27 October 2026

Abstract

We replace the elliptic shift by an auxiliary Gamma driver while retaining the full variable-metric spherical Z4c equations. Algebraic elimination enforces the normal nullness relation at every Runge–Kutta stage and accepted state. Across the initial 31 bounded runs the maximum relative nullness error is one binary64 machine epsilon. A source-restored candidate exhibits approximately second-order spatial self-convergence and fourth-order timestep convergence in short regular-center tests. Subsequent three-grid puncture tests reach $3M$ with decreasing errors, but the original duration extensions fail near $8M$. An interior-only source adaptation postpones failure to 16–17M. A subsequent constant-strength bulk Gamma driver, with an independently audited inner gauge stencil, reaches $24M$ on two grids; three-grid tests through $3M$ improve under refinement. A compact lapse-adjusted Z4 damping control subsequently reaches $30M$ on the coarse grid with positive metric and small horizon-mass drift, but a later extension fails at $45.525M$ after exterior lapse/shift speed coincidences develop. Doubling the constant coefficient delays the coarse failure to $74.7M$ and gives positive short spatial and temporal refinement, but a transient speed coincidence recurs. A fresh coefficient-four control completes $100M$ on the coarse grid; matched three-grid spatial refinement extends through $12.6M$, and independent timestep tests through $9M$. Regional diagnostics reveal lower orders near the puncture and scri than in the exterior aggregate. Single-grid survival does not yet qualify the persistent three-grid $100M$ stability and convergence target. Exact nullness therefore succeeds as a discrete gauge identity without establishing full boundary compatibility, puncture regularity, or improved robustness over the elliptic formulation.

1 Preserved baseline and proposed driver

The reference is commit `a9214bccf` of the isolated elliptic-shift checkout. Its tracked state, jobs, and historical evidence were left untouched. The new branch is `codex/hyperboloidal-hyperbolic-shift`. The reference $N = 1025$ wormhole evolution reached coordinate $t = 500$ with final relative apparent-horizon mass error 1.14804×10^{-6}; that single-grid result is a duration certificate, not a convergence proof.

Use $\Omega = (1-r^2)/2$, $A = (1+r^2)/2$, $u = \alpha - A$, $p = K_{\rm phys} - 2\Theta_{\rm phys}$, and $\mathcal{D}_t = \partial_t - \beta\partial_r$. The twice-conformal spatial metric has components $(g, r^2 g^{-1/2}, r^2 g^{-1/2})$. Both g and the independent Z4 connection Λ^r evolve. The metric-built connection is

$$V^r = \frac{g'}{g^2} + \frac{2}{r}(\sqrt{g} - g^{-1}), \qquad Z_r = \frac{g}{2}(\Lambda^r - V^r). \tag{1}$$

It is a diagnostic; Λ^r is never replaced by V^r in the bulk.

The retained lapse and relaxation flux are

$$\mathcal{D}_t\alpha = r^2 + \frac{-F(p+3) + r(\alpha\beta + r) - 2\xi\alpha u + 16\Omega^4 r^2 u/A}{\Omega} + \eta\chi\left[wB' + (w' + 2w/r)B\right], \tag{2}$$

$$\tau\mathcal{D}_t B = \kappa w u' - B, \qquad F = \alpha^2 + 4\Omega^2\alpha, \quad \eta = \frac{(1+\epsilon)\alpha}{\alpha + \epsilon A}. \tag{3}$$

Here $\kappa = 0.005$, $\tau = 0.02$, $\epsilon = 10^{-4}$, and $\xi = 1 + 2(1 - r^2)^4$. The baseline window w equals one at the puncture and vanishes for $r \geq 0.8$, with transition width 0.24. The attenuation multiplies the flux outside its divergence. The disclosed Appendix-C scalar-Θ source modification and baseline Z4 damping $(\kappa_1, \kappa_2) = (4, -0.99)$ are retained, except in the explicitly labelled interior-damping control below.

With a separate shift auxiliary D, the candidate is

$$\mathcal{D}_t\beta = -r + \mu D - \xi_\beta\frac{\beta + r}{\Omega}, \tag{4}$$

$$\mathcal{D}_t D = \mathcal{D}_t\Lambda^r - \eta_D D, \qquad \eta_D = 1, \tag{5}$$

$$\mu = \frac{3}{4}\lambda\chi\left(F + g\eta\kappa w^2/\tau\right), \qquad \lambda = 2. \tag{6}$$

The analytic source $-r$ cancels the reference shift advection at Minkowski equilibrium. No numerical Einstein-equation RHS is subtracted. The first candidate used $\xi_\beta = 0$; the stronger tested candidate uses $\xi_\beta = 4$. This is a hyperboloidal adaptation of the auxiliary Gamma-driver family discussed in Vañó-Viñuales and Husa (2018), with a different speed choice and exact algebraic nullness closure.

2 Exact nullness and compatible boundary variables

Let $n_i = -\partial_i\Omega$ point outward, since Ω is positive inside the domain, and let $\bar{\gamma}_{ij} = \tilde{\gamma}_{ij}/\chi$ be the regular spatial metric. With $\partial_t\Omega = 0$,

$$\bar{g}^{ab}\Omega_a\Omega_b = \bar{\gamma}^{ij}n_i n_j - \frac{(\beta^i n_i)^2}{\alpha^2} = 0. \tag{7}$$

Defining $s_i = n_i/(\bar{\gamma}^{jk}n_j n_k)^{1/2}$ gives the future branch $\beta^i s_i = -\alpha$. A boundary representation is

$$\beta^i = \beta^i_T - \alpha s^i, \qquad s_i\beta^i_T = 0. \tag{8}$$

Nullness determines one normal component. Tangential coordinate choices remain separate. For fixed Ω_i, the moving-normal rate is

$$\partial_t s_i = \frac{1}{2}s_i s^k s^l\partial_t\bar{\gamma}_{kl}, \qquad s_i\partial_t\beta^i = -\partial_t\alpha - \beta^i\partial_t s_i. \tag{9}$$

Thus evolving the metric cannot be omitted when differentiating the condition.

In spherical symmetry the dependent shift endpoint is reconstructed directly:

$$\beta_1 = -\alpha_1\sqrt{\chi_1/g_1}, \qquad \dot{\beta}_1 = \beta_1\left(\frac{\dot{\alpha}_1}{\alpha_1} + \frac{\dot{\chi}_1}{2\chi_1} - \frac{\dot{g}_1}{2g_1}\right). \tag{10}$$

The lapse clock is $\alpha_1 = 1$, $\dot{\alpha}_1 = 0$. Every stage reconstructs β_1 before spatial derivatives, and every accepted state is reconstructed again. The differentiated relation alone would not enforce the discrete identity.

Writing $v = \beta + r$, the regular limit of the restoring term is $+\xi_\beta v_1'$ when $v_1 = 0$. The same boundary datum determines the auxiliary trace through the first driver equation:

$$D_1 = \frac{\dot{\beta}_1 - \beta_1\beta_1' + 1 - \xi_\beta v_1'}{\mu_1}. \tag{11}$$

There is no second independently prescribed value for D. Its endpoint storage slot is eliminated. Interior D and Λ^r remain independent. This changes a gauge boundary closure; it does not project the Hamiltonian, momentum, or Z4 constraints. The derivative stencil in Eq. (11) is the same one used in the shift equation.

All geometric equations are reevaluated using the unreduced tensor kernel, as $q_t = R_q + S_q/\Omega$. At actual scri the candidate uses $R_q - S'_q$, with S' obtained from the complete variable-metric numerator. Neither the elliptic relation for β' nor its $g = 1$ endpoint equations are substituted. No floor is applied to Ω.

The retained weak regularity residuals are

$$\mathcal{C} = \left(\chi - g,\ 0,\ A_{rr} - g'/6 + 2\chi'/3,\ p + 3,\ \Theta,\ \Lambda^r - V^r,\ 0\right). \tag{12}$$

They select the same clock-compatible asymptotic phase, allowing the endpoint value of g to evolve. The endpoint rates receive $4S_q/h_{\text{end}} - 8\mathcal{C}_q/h_{\text{end}}$. The restoring numerator $-\xi_\beta v$ is monitored separately. These traces do not enforce the entire derivative hierarchy: on this phase, for example, $S_\Lambda = 0$ also requires $2p' + \Theta' = 6A_{rr}/g$. Neither all numerators nor the raw lapse-clock rate are imposed exactly. The clock override and finite penalties therefore still require a stability and compatibility assessment.

3 Characteristic audit and its limits

An interior first-order normal reduction uses

$$U = (\chi', g', A_{rr}, \delta p/\Omega, \Theta/\Omega, \Lambda^r, \alpha', \beta', B', D). \tag{13}$$

For $\mathcal{D}_t U = P\partial_r U$, exact symbolic factorization gives

$$\det(zI - P) = z^2(z^2 - c_P^2)^2(z^2 - c_L^2)(z^2 - c_S^2), \tag{14}$$

where

$$c_P^2 = \alpha^2\chi/g, \quad c_L^2 = F\chi/g + \eta\chi\kappa w^2/\tau, \quad c_S^2 = 4\mu/(3g) = \lambda c_L^2. \tag{15}$$

Coordinate speeds are $-\beta$, twice, together with $-\beta \pm c_P$, each twice, $-\beta \pm c_L$, and $-\beta \pm c_S$. Native tensor coefficient checks agree with this scalar symbol to 8.4×10^{-12} or better.

At normalized scri $(\alpha, \chi, g, \beta) = (1, 1, 1, -1)$ and $w = 0$. For $\lambda = 2$ the scalar reduction has one incoming mode, six outgoing modes, and three tangent modes. Exact eigenspaces are complete there. The incoming shift mode motivates one normal gauge datum. For $\lambda < 1$ there is no incoming shift mode; adding an independent Dirichlet datum would require a different compatibility argument. For $\lambda = 1$, the light-speed eigenvalues each have algebraic multiplicity four but geometric multiplicity three: this driver choice is defective and was not adopted.

The real speeds in Eq. (14) alone are not a proof of strong hyperbolicity. Numerical positive-geometry samples and exact normalized-scri eigenspaces support the scalar candidate, but a uniform symmetrizer for arbitrary geometry has not been constructed. The p/Ω reduction is itself singular at scri, so its finite limiting matrix does not prove well-posedness of the actual boundary equations. At the exact collapsed limit $\alpha = \chi = 0$, the displayed matrix is nonzero and nilpotent of index two, hence defective. No uniform puncture estimate follows from a positive lapse floor. The full three-dimensional vector and tensor sectors, including a direction-independent telegrapher extension, remain to be qualified.

4 Numerical implementation and short-time evidence

The complete spherical tensor equations evolve g, A_{rr}, p, Θ, and Λ^r freely. Only determinant and trace-free algebraic identities are built into the spherical tensor representation. Seven-point finite differences, biased

advection, KO8 strength 0.5 with the baseline radial taper, and classical RK4 are used. Regular-center tests use half cells at the center and an actual scri node. Their KO8 acts on stored deviations; it is distinct from the historical regular-center KO6 filter on rescaled tensor coefficients. The matched elliptic controls use the same full tensor kernel and numerical settings. Their coordinate profiles need not match those of the hyperbolic gauge.

The finite lapse pulse is a compact smooth bump of amplitude 0.01, centered at $r = 0.4$ with half-width 0.12. With $\xi_\beta = 4$, grids $N = 65, 129, 257$ reach $t = 1$. The mixed-field self-differences, interpolated with seven-point polynomials to the coarse grid, are 1.38688×10^{-5} and 3.42764×10^{-6}, giving approximate order 2.02. This is an observed short-test order, not sixth-order accuracy of the complete scheme. At fixed $N = 65$ and $t = 0.15$, halving the timestep twice gives differences 8.22131×10^{-10} and 5.12329×10^{-11}, or order 4.00. Boundary and bulk accuracy are not interchangeable.

Gauge and resolution at $t = 1$	$\|H\|_{r^2dr}$	$\|M_r\|_{r^2dr}$	scri pole
Hyperbolic, $N = 65$	4.31×10^{-4}	7.08×10^{-4}	3.22×10^{-5}
Hyperbolic, $N = 129$	2.47×10^{-4}	3.54×10^{-4}	1.21×10^{-5}
Hyperbolic, $N = 257$	7.80×10^{-5}	9.23×10^{-5}	5.07×10^{-6}
Elliptic, $N = 129$	2.23×10^{-5}	3.51×10^{-5}	8.27×10^{-8}

The new gauge has larger constraint errors than the matched elliptic control. Radial-weighted constraint norms are useful consistency diagnostics, but their finite-error values are gauge dependent. Apparent-horizon mass supplies the invariant puncture comparison below.

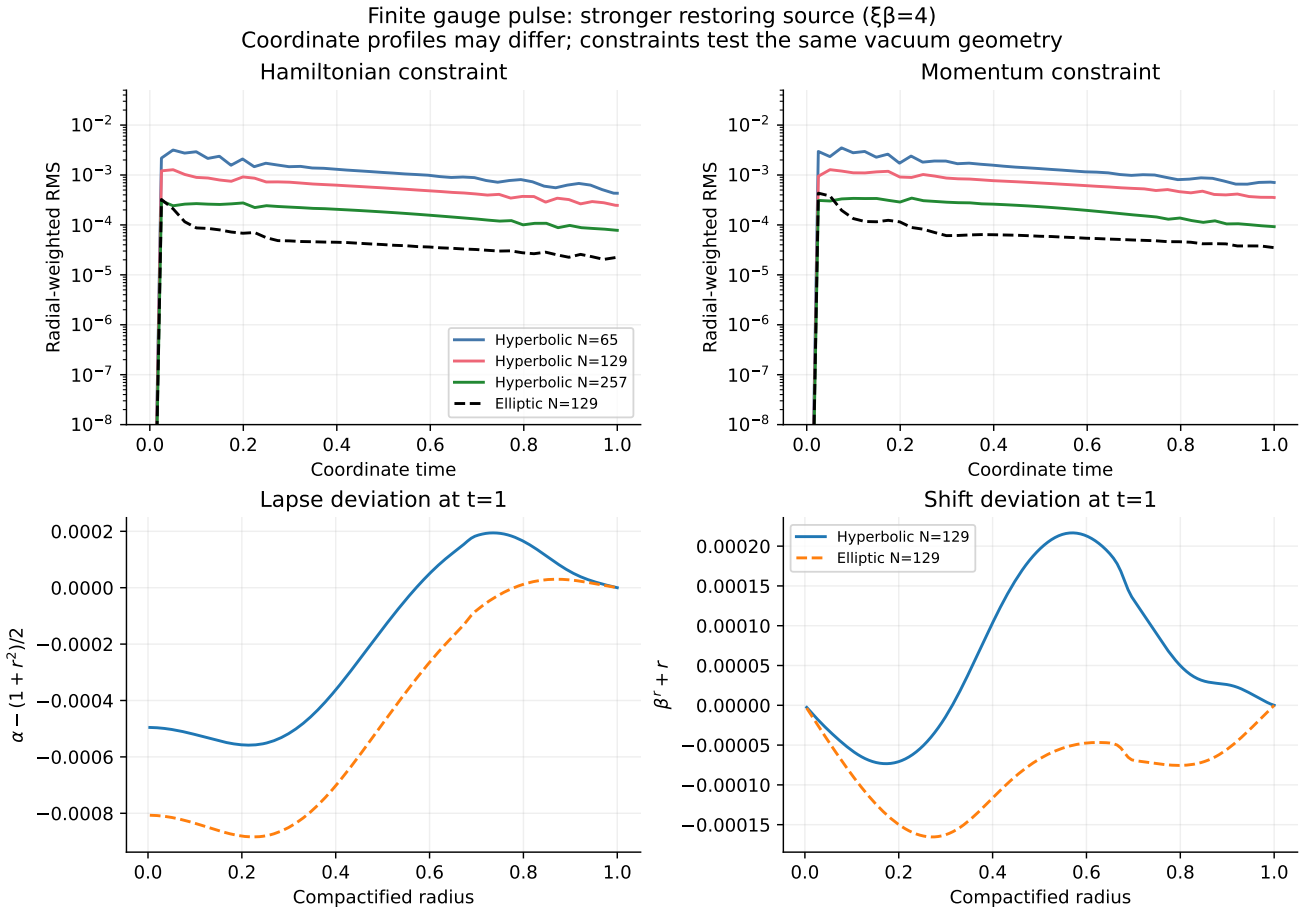


Figure 1: Finite pulse with $\xi_\beta = 4$. Constraint errors decrease under refinement. The lower panels illustrate different coordinate responses, without interpreting their difference as physical error.

5 Failures, puncture treatment, and qualification decision

Without restoring, the $N = 65$ pulse violates the predeclared exploratory scri-numerator threshold 10^{-2} at $t = 0.248062$. Varying the speed ratio, increasing auxiliary damping, or halving KO strength did not remove that failure. With $\xi_\beta = 1$, $N = 65$ and 257 reach $t = 1$, but $N = 129$ fails at $t = 0.249027$. This nonmonotonic behavior is preserved rather than called convergence. The stronger source succeeds in the regular-center refinement sequence above.

Linearized method-of-lines spectra at Minkowski, $N = 33, 65$, have negative spectral abscissae for the three tested restoring strengths. Thus these spectra do not explain the nonlinear pulse failures as an already demonstrated exponentially growing linear eigenmode. Nonnormal amplification, nonlinear boundary compatibility, and finite-resolution coupling remain possible mechanisms. Exact nullness is maintained even in every failed trajectory.

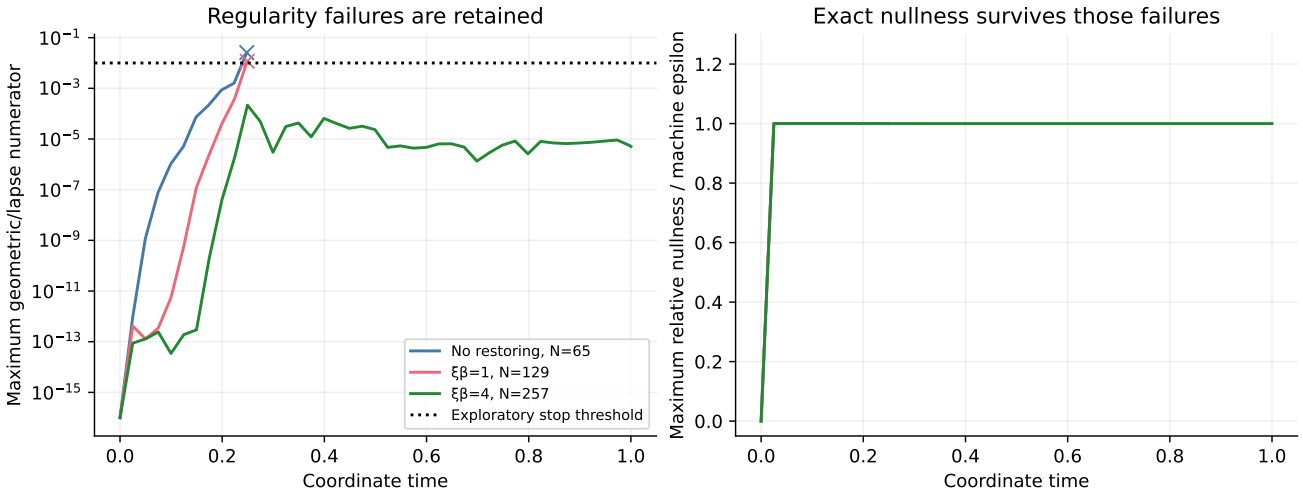


Figure 2: Failed controls and a surviving regular-center candidate. Crosses mark actual stop states. The nullness diagnostic accumulates over all RK stages and accepted states; it does not imply cancellation of the other scri numerators.

The wormhole trials preserve $M = 1/3$ and the baseline matched isotropic initial slice. They retain the radial map with stretch 16, positive-node stencils, reference-weighted positive variables, and physical lapse floor 10^{-10}. All positive-radius nodes evolve; none are excised. The RK lapse variable is the total conformal lapse. Accepted increments use compensated accumulation. A separate total-value tensor adapter avoids subtracting small lapse, conformal factor, and physical trace from reference values. The initial shift is matched to the elliptic datum; subsequent stages use only the local driver.

The coefficient endpoint is rederived for $g = 1 + g_1 r + \cdots$ and $\beta = b_1 r + b_2 r^2 + \cdots$. In particular,

$$\partial_t \Lambda_0 = 2b_1\Lambda_0 - 4b_1 g_1 + \frac{16}{3} b_2, \qquad \partial_t g_1 = b_1 g_1 + \frac{4}{3} b_2. \tag{16}$$

The leading logarithmic χ and lapse rates remain $2b_1$ and $2b_1 - 6 + 2\xi(0)$. Independent tensor evaluations approaching $r = 0$ converge to these coefficient limits. This checks the leading expansion, not a uniformly accurate puncture evolution.

Fresh trials use $\xi_\beta = 4$ and $\Delta t \le 0.0125h$. At $N = 129$ the scri guard stops the evolution at $t = 0.19423828125$; $N = 257$ reaches its short target $t = 0.2$. The final relative errors with respect to analytic M are 7.7463×10^{-4} and 3.9726×10^{-6}, respectively. Maximum mass changes relative to each numerical initial horizon mass are 1.1975×10^{-3} and 1.2846×10^{-5}. Two unequal-duration grids do not establish puncture convergence or wormhole-to-trumpet relaxation.

For exterior constraints, the mask uses the instantaneous apparent horizon and the RMS is normalized by the sum of exterior radial quadrature weights. The full interior diagnostics are retained separately. At the final saved states the interior maximum $|H|$ is 0.449 and 0.00164. The largest used-minus-original rate differences are in Λ^r near $r = 3.65 \times 10^{-8}$ and 1.60×10^{-8}, of order 4.64×10^4 and 2.01×10^4. These include the finite-difference advection and dissipation modifications, not an independent continuum residual. They prevent an accuracy claim near the puncture. The floor activates only inside $r \simeq 6.29 \times 10^{-7}$.

The independently evaluated raw lapse-clock rates at the final endpoints are 0.0276 and -0.00438, despite the imposed $\alpha_t = 0$. These compatibility defects, together with the scri numerators, are a reason to revisit the boundary closure. They do not identify a unique cause of failure. The later bounded duration probes below reassess this short-test decision; no 3D run has been launched.

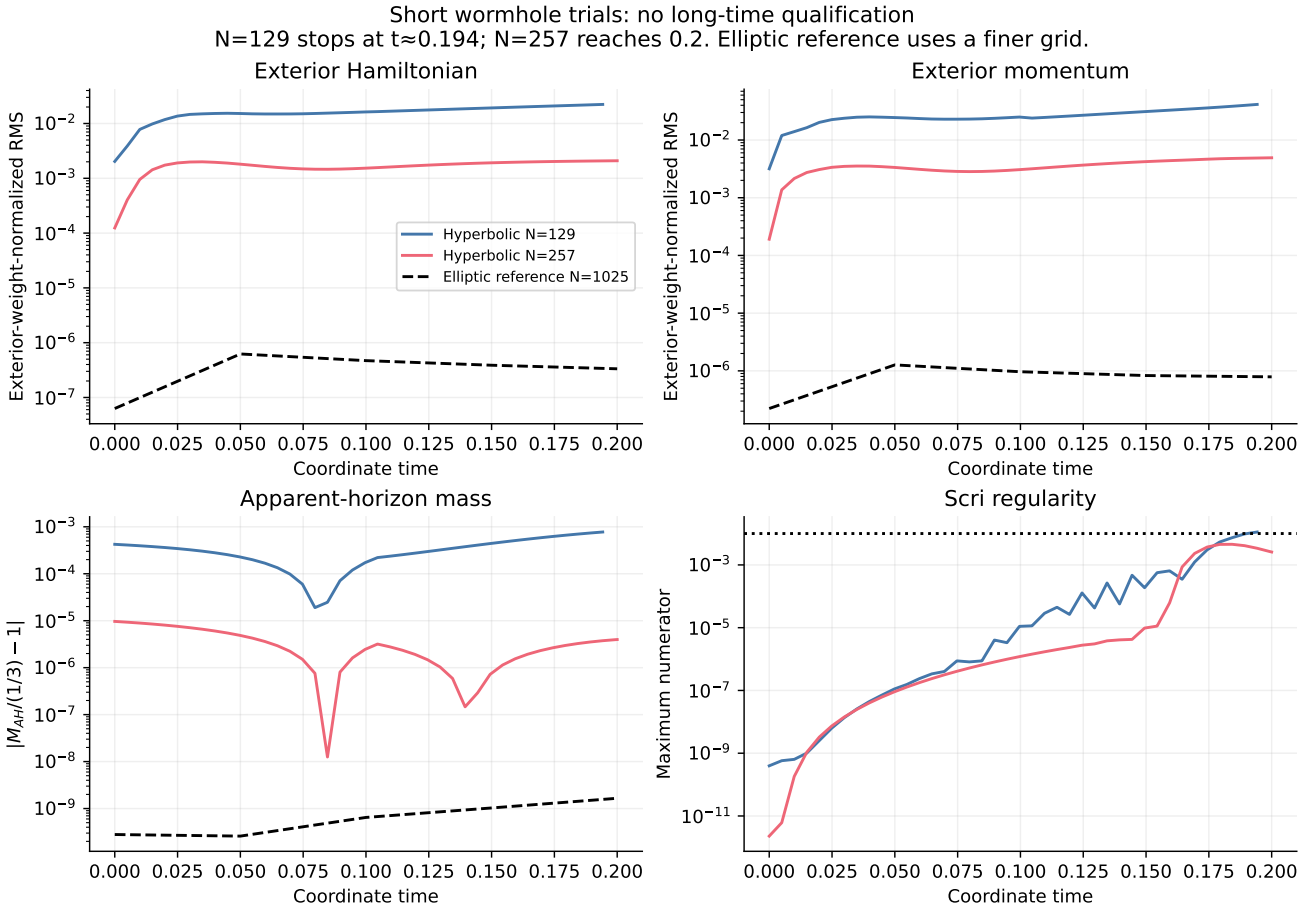


Figure 3: Short puncture tests. The dashed elliptic reference is $N = 1025$, so this is not a matched-resolution superiority comparison. Mass errors include the initial discretization error; changes from the numerical initial mass are reported separately in the text.

6 Bounded extensions toward the persistent $100M$ target

The duration target is $t/M = 100$, or coordinate $t = 100/3$ for the retained $M = 1/3$. The goal remains active. The predeclared criteria require three nested grids to reach that duration, decreasing exterior constraints and mass errors, positive-order field self-convergence, a separate timestep check, and continued positivity on every positive-radius node. A small horizon-mass error alone cannot certify a

usable evolution.

At coordinate $t = 1 = 3M$, grids $N = 257, 513, 1025$ all complete with $\Delta t = 0.05h$. Comparing $\Delta t = 0.05h$ against $0.0125h$ at $N = 257$ changes the final conformal lapse by at most 3.5×10^{-13} and the horizon mass by 2.1×10^{-14}. This bounds the measured time sensitivity; two stepsizes do not establish a new temporal order.

N	exterior H RMS	exterior M_r RMS	$\lvert M_{\mathrm{AH}}/M - 1\rvert$	peak scri numerator
257	9.49×10^{-3}	1.86×10^{-2}	2.92×10^{-5}	4.90×10^{-3}
513	9.81×10^{-4}	9.12×10^{-4}	1.10×10^{-6}	2.99×10^{-3}
1025	4.33×10^{-4}	5.37×10^{-5}	5.81×10^{-8}	1.77×10^{-3}

The full radial-weighted mixed-field self-differences are 1.7606×10^{-2} and 1.3080×10^{-2}, giving only order 0.43 at this time. The independently evolved connection dominates this mixed norm. These are short-time refinement results, not a satisfactory long-time convergence certificate. No puncture neighborhood is removed from the profile comparison.

Two fresh extensions target coordinate $t = 3 = 9M$. The $N = 257$ run stops at $t = 2.825$ when its scri numerator exceeds 10^{-2}. Its interior Hamiltonian maximum has already reached 3.19×10^4 near $r = 0.0672$. The $N = 513$ run fails earlier, at last accepted time $t = 2.726953125$, because the next RK stage produces $g = -5.91 \times 10^{-6}$ near $r = 0.09365$. A one-step postmortem replay reproduces that failure. The finer run's scri numerator remains below 0.0051, so raising the scri guard would not cure this failure. The last scheduled diagnostics precede a between-sample stage failure; manifests and terminal arrays retain their separate exact times.

Both runs retain small horizon-mass drift: the largest sampled changes relative to their numerical initial masses are 2.77×10^{-4} and 1.07×10^{-5}. Nevertheless, the loss of positive spatial metric makes the second run unusable. The failing region is far outside the lapse-floor radius, at most 7.32×10^{-6}. An outer-collar Jacobian diagnostic has negative spectral abscissae at the sampled times, but its artificial inner interface and exclusion of the bulk prevent a full-domain stability conclusion.

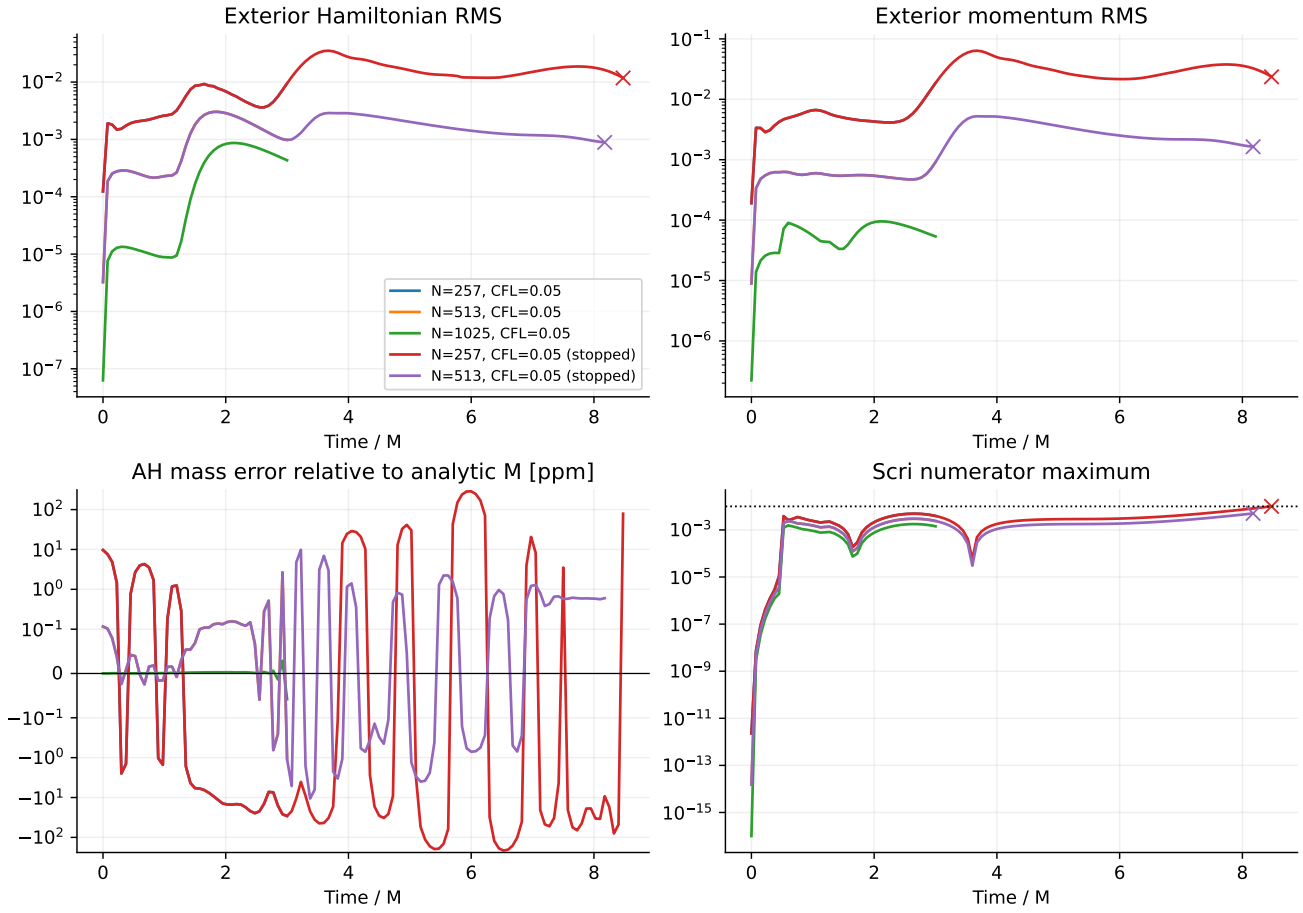


Figure 4: Duration probes with the original hyperbolic driver. Curves terminate at their actual saved diagnostic times; crosses identify failed runs. The $N = 1025$ trajectory is only a $3M$ probe. The horizon-exterior norms decrease with refinement, while interior metric breakdown still prevents continuation toward $100M$.

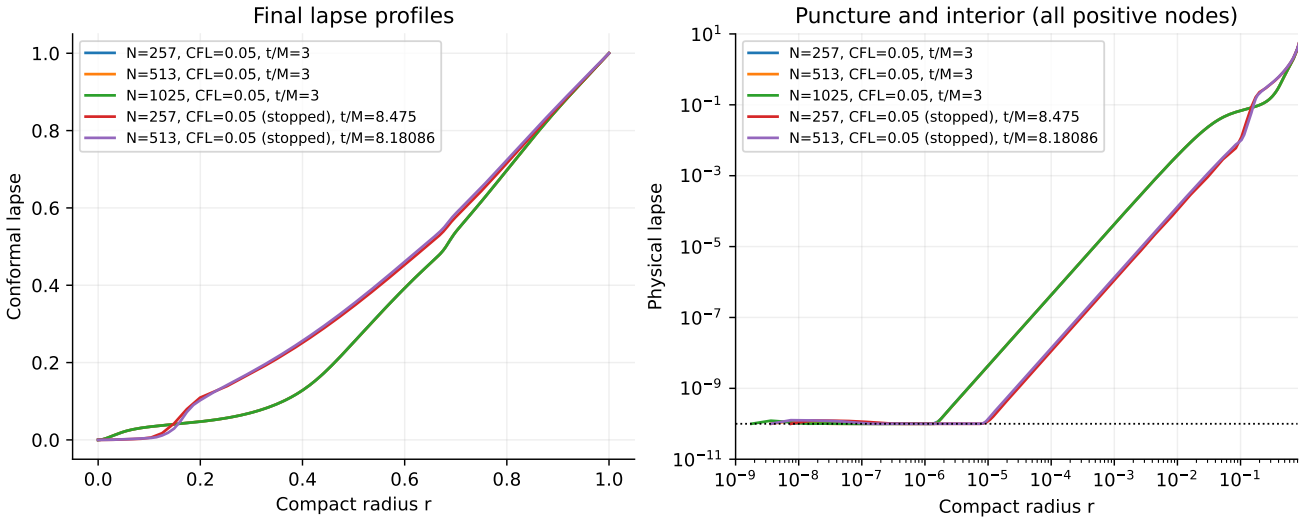


Figure 5: Lapse snapshots at the indicated times, retaining the telegrapher coupling and physical floor 10^{-10}. Different terminal times must not be interpreted as a convergence comparison. The floor region is several orders of magnitude closer to the puncture than the observed metric failure.

A bounded candidate adds $b\Omega^2\sqrt{\chi}$ to μ, with $b = 0.1$. It leaves the scri coefficient and incoming-mode count unchanged and gives

$$c_S^2 = 2c_L^2 + \frac{4b\Omega^2\sqrt{\chi}}{3g} > 2c_L^2 \quad (\Omega > 0,\ \chi, g > 0). \tag{17}$$

Before lapse flooring changes the leading AF scaling, the shift speed changes from $O(r^3)$ to $O(r)$. This is a gauge-transport hypothesis, not a cure established by the symbol. Three regular-center grids pass the same short gate, but the fresh $N = 257$ puncture trial fails at $t = 2.8078125$ when a stage metric becomes negative near $r = 0.0672$. This candidate therefore does not resolve the duration failure and is not the default.

A mechanism tested next is the collapsed-region background forcing. For smooth AF data with $\beta = b_1 r + O(r^2)$, the original driver gives, at continuum leading order,

$$\dot{b}_1 = b_1^2 - 1 - 2\xi_\beta(b_1 + 1). \tag{18}$$

At $\xi_\beta = 4$ this has a stable fixed point $b_1 = -1$. Thus the driver actively restores the inward reference shift even in the collapsed-lapse region. The added $b\Omega^2\sqrt{\chi}D = O(r^2)$ does not change this leading forcing. This motivates the controlled source changes below; an increased survival time alone does not establish a unique cause of the original failure.

The implementation now writes atomic progress records and bounds diagnostic spacing by $\Delta t_{\text{report}} \le 0.025$. A diagnostic-only optimization omits constraint and horizon calculations during intermediate RHS evaluations while retaining them at reports; the native rate arrays agree bit for bit with diagnostics enabled. Full failed-stage arrays are preserved for newly instrumented failures. The original failed runs and elliptic reference remain unchanged.

6.1 Lapse-dependent shift sources and a metric-connection control

The telegrapher lapse, its $\eta\chi$ attenuation, and the physical lapse floor remain unchanged in every control here. We replace only the shift background terms by

$$\mathcal{D}_t\beta = -r\sigma^2 + \mu D - \xi_\beta\frac{\beta + r\sigma}{\Omega}. \tag{19}$$

The global choice is $\sigma = \alpha/A$. Its complete restoring limit is $\xi_\beta[\beta' + (r\sigma)']$; the σ' term must be included when eliminating the dependent D trace. Algebraic nullness still holds at each stage, but the $N = 257$ global-source trial fails its scri guard at coordinate $t = 0.175$. We retain that failure.

The interior-only choice is $\sigma = 1 + w(\alpha/A - 1)$, using the existing smooth window. It equals α/A near the puncture and exactly one for $r \ge 0.8$. Hence the original source and boundary closure are unchanged throughout the outer collar. The change is algebraic in the evolved fields, so the interior principal matrix is unchanged. Before flooring alters the AF expansion, the leading slope equation becomes

$$\dot{b}_1 = b_1^2 - 2\xi_\beta b_1, \tag{20}$$

whose stable fixed point is zero. Independent tests verify the leading AF rate and full scri limit, the unperturbed Minkowski equilibrium, and roundoff nullness for perturbed stage data.

Three regular-center pulse grids pass their short gate with decreasing constraint errors. Fresh puncture grids $N = 257, 513$ both reach $t = 3 = 9M$. Their exterior Hamiltonian RMS values are 9.91×10^{-3} and 4.02×10^{-4}; their absolute relative horizon-mass errors are 1.81×10^{-3} and 7.42×10^{-5}. These two-grid improvements do not establish convergence of the modified gauge through $100M$.

Both trajectories continue from verified checkpoints through $t = 5 = 15M$. The subsequent extensions fail at $t = 5.3609375$ and 5.6947265625, respectively. A stage metric becomes negative near $r = 0.00332$ on the coarse grid and $r = 0.01156$ on the finer grid. The fine trajectory's maximum sampled horizon-mass drift is only 2.03×10^{-4}, and its last sampled exterior Hamiltonian RMS is 1.51×10^{-3}. These exterior diagnostics do not make a state with invalid spatial geometry evolvable. Interior constraint magnitudes are diagnostic evidence; the stopping condition is loss of metric positivity, not an interior error budget.

A separate gauge control sources D from the metric-derived connection V, while continuing to evolve Λ^r independently in the unchanged Z4 system:

$$\mathcal{D}_t D = \mathcal{D}_t V - \eta_D D, \qquad G = \mathcal{D}_t g = -2\alpha A_{rr} + \frac{4g}{3}(\beta' - \beta/r), \tag{21}$$

$$\mathcal{D}_t V = \frac{G' + \beta' g'}{g^2} + \left[-\frac{2g'}{g^3} + \frac{g^{-1/2} + 2g^{-2}}{r} \right] G + \frac{2\beta}{r^2}(\sqrt{g} - g^{-1}). \tag{22}$$

This is a local hyperbolic gauge source, not an elliptic solve or a replacement of Λ^r by V in the Einstein equations. The last principal-symbol row becomes $(-2\alpha/g^2)\partial_r A_{rr} + [4/(3g)]\partial_r^2 \beta$. Exact factorization yields the same scalar polynomial as Eq. (14), with complete normalized-scri eigenspaces for $\lambda = 2$ and the same degeneracy at $\lambda = 1$. The full puncture and 3D well-posedness limitations remain.

A complex-step chain-rule oracle checks the metric-connection rate to 2.1×10^{-16}, and the metric rate agrees with the native tensor kernel to 5.6×10^{-17} in the tested states. The implementation evaluates this continuum rate with central derivatives and gives D its own biased transport and KO terms. Unlike the evolved-Λ control, it does not feed numerical dissipation of Λ into D. It passes the regular-center gate and the $N = 257$ puncture test through $9M$, but subsequently loses metric positivity at $t = 5.487109375$ $(16.46M)$. It is therefore not a duration repair.

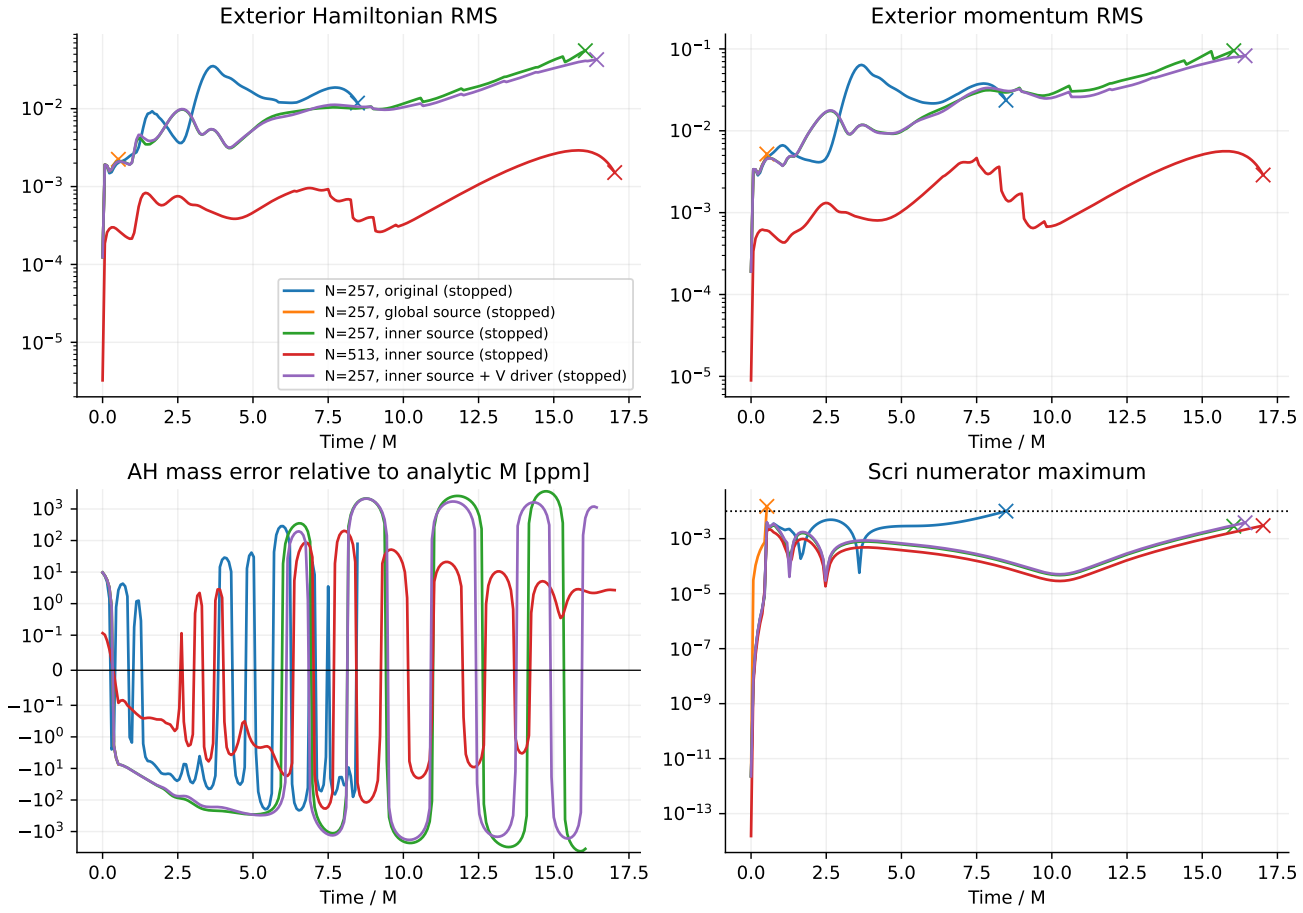


Figure 6: Failed source/driver controls, all retaining the telegrapher lapse. Each curve includes its verified continuation history. Crosses show the last saved diagnostics of a failed trajectory; exact last-accepted and failed-stage times are retained separately. The finer interior-source run has much smaller exterior errors but still fails before $18M$.

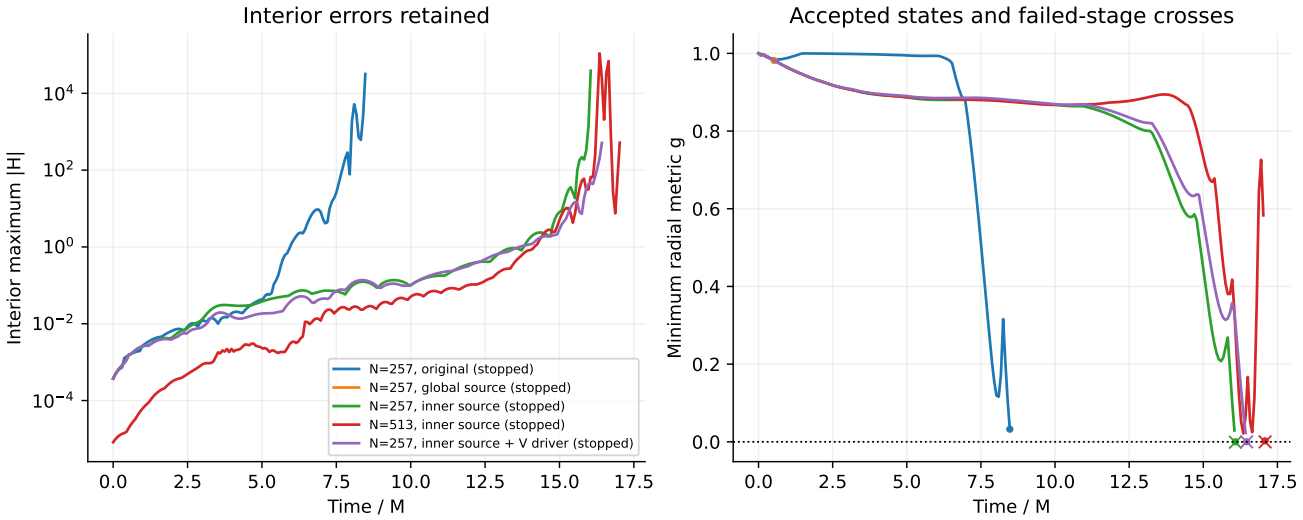


Figure 7: Interior errors and metric positivity. Filled dots add the last accepted metric minimum; crosses in the metric panel use the actual failed-stage array when available. The global-source trial stops on its scri guard before developing the later interior failure. No interior nodes are removed from the evolution.

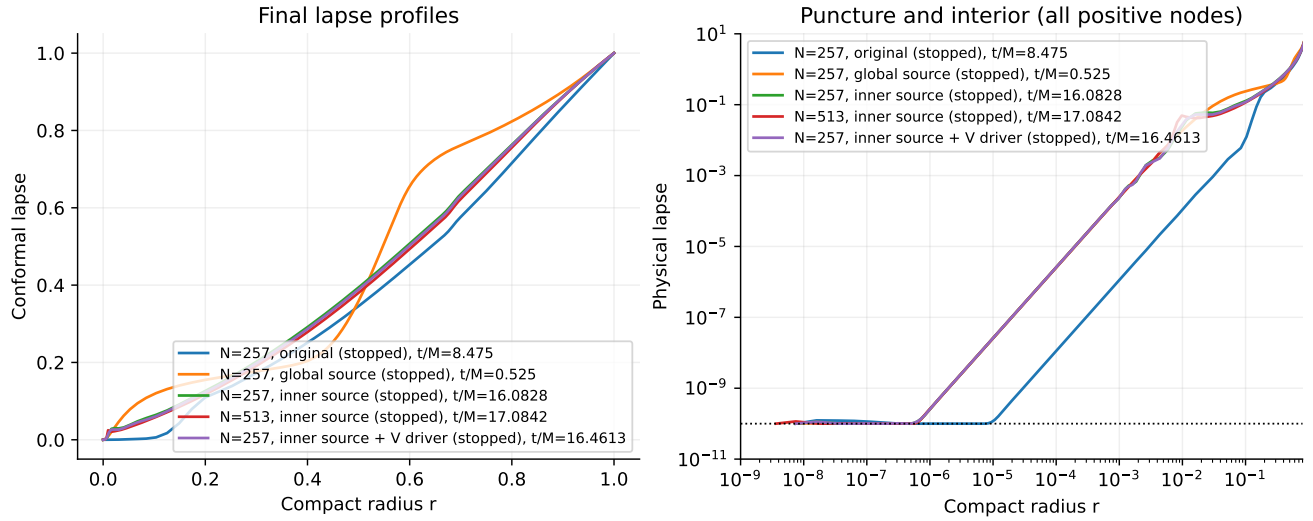


Figure 8: Final lapse snapshots for the same failed controls, at unequal times as labeled. The physical floor is retained. These profiles cannot serve as a common-time convergence test.

The continuation harness accepts completed parents only, verifies source, library, checkpoint, and parent-record hashes, loads the frozen parent equations, and inherits the exact timestep and RK compensation. Two-generation split/uninterrupted tests agree bit for bit. A test initially exposed a one-ulp difference caused by recomputing $\Delta t = t_{\rm end}/N_t$; fresh runs now preserve the nominal CFL h bit pattern when the target is on that lattice. Corrupt histories, changed sources, failed parents, and unaligned continuation targets are rejected. These short restart tests establish implementation consistency, not long-time physical accuracy.

A possible later numerical control is to use $\log g$ and the mixed curvature A_{rr}/g as evolution variables. That change would require a separate chain-rule/native-kernel audit, including loss of precision in the current deviation-based tensor adapter when g becomes very small. No metric floor or logarithmic-metric evolution has been used. The requested next direction instead restores a moving-puncture-like shift in the bulk, as described below. The $100M$ objective remains unmet.

6.2 An unsuppressed bulk Gamma driver

The next candidate follows the fully advective two-variable Gamma-driver structure in Eqs. (18) and (20) of van Meter et al. (2006). Its shift coefficient does not vanish with the lapse or conformal factor. Let $C(r)$ be zero for $r \le 0.8$, one for $r \ge 0.9$, and $x^8/[x^8 + (1-x)^8]$ in between, where $x = (r - 0.8)/0.1$. We use

$$\mathcal{D}_t \beta = \mu_\Gamma D - r - 4C(r)\frac{\beta + r}{\Omega}, \tag{23}$$

$$\mathcal{D}_t D = \mathcal{D}_t \Lambda^r - D, \qquad \mu_\Gamma = (1-C)\mu_{\rm bulk} + C\mu_{\rm separated}. \tag{24}$$

Here $\mu_{\rm separated}$ is the earlier coefficient with $\lambda = 2$. Thus the bulk has a constant-strength advective Gamma driver and no reference-restoring force. The prescribed $-r$ source remains necessary to preserve the chosen hyperboloidal Minkowski equilibrium with $D = 0$; this is a disclosed difference from the asymptotically flat gauge. No measured Einstein-equation residual is subtracted. The outer collar recovers the previous driver and its complete scri limit exactly. The algebraic normal shift and dependent auxiliary trace are imposed at every stage as before; no additional scri datum is supplied.

The coefficient is configurable. The conventional $\mu_{\rm bulk} = 3/4$ is not used for the first evolution: with the retained telegrapher lapse at the Minkowski center, $c_L^2 = 1 = 4\mu_{\rm bulk}/3$. Exact rational eigenspace calculations give algebraic multiplicity two but geometric multiplicity one for each of the speeds ± 1. Setting $\mu_{\rm bulk} = 1$ separates those speeds and gives complete scalar eigenspaces at that state. The general polynomial remains Eq. (14), with the new μ_Γ. At normalized scri there are still one incoming, six outgoing and three tangent scalar modes. At the exactly collapsed corner $\alpha = \chi = 0$, the zero eigenvalue has algebraic multiplicity eight and geometric multiplicity seven even with this unsuppressed driver. No uniform collapsed-lapse or full 3D strong-hyperbolicity claim follows.

For $N = 257$, the original stretch-16 map would give a mapped gauge speed of 5.66×10^5 and a Courant number of 2.83×10^4 at $\Delta t/h = 0.05$. The new puncture controls therefore explicitly set the map strength to zero, retain the C7 outer grid collar, and use $\Delta t/h = 0.025$. The first positive radius is then $1/256$, rather than 7.49×10^{-9}. This is a different grid family; improved duration on it cannot isolate the gauge change from the mesh change.

The implementation audits confirm an exact bulk shift-rate identity, unchanged native Einstein and telegrapher-lapse rates at identical states, independent evolution of Λ^r, and stage nullness below 2.3×10^{-16}. The regular-center Minkowski residual is 1.47×10^{-13}. Three finite-pulse grids complete through $t = 1$, with mixed-field self-differences 2.374×10^{-3} and 6.252×10^{-4}, corresponding to an approximate order of 1.925. An independent fixed-grid pulse test at $t = 0.15$ uses $\Delta t = 0.0015, 0.00075, 0.000375$ and gives temporal order 4.004. This is a short regular-center result, not long puncture temporal convergence.

The first puncture implementation retained positive-node extrapolation for shift derivatives. Its coarse $N = 257$ run reached $t = 0.2$, but $N = 513$ lost metric positivity at $t = 0.149462890625$, at the third positive node. Halving the timestep gives a failure at $t = 0.1494873046875$, again at the third positive node. The near-identical failure time and reversal under spatial refinement reject that implementation as a stable candidate. A separate reduced radial-vector wave test isolates the leading gauge equation,

$$\partial_t^2 \beta + \partial_t \beta = \frac{4\mu_{\rm bulk}}{3} \left(\beta'' + \frac{2\beta'}{r} - \frac{2\beta}{r^2} \right), \tag{25}$$

about the collapsed, zero-shift leading state, with the same KO8 coefficient and four fixed outer nodes. This isolates the singular inner spatial closure; background advection and the complete coupled geometry are omitted. Positive-node extrapolation gives maximum growth rates 26.16 and 52.68 on $N = 65, 129$. An odd radial-vector extension gives maximum real eigenvalues approximately -0.5 on both grids. This reduced spectral test identifies a closure instability; it is not a stability proof for the complete Einstein system.

The resulting control uses odd extension only for shift and auxiliary derivatives. At the fixed spherical puncture, $\beta(0) = 0$ implies $D(0) = 0$ through the unsuppressed shift equation; the auxiliary value is dependent, not an extra independent gauge datum. The AF geometry, lapse, curvature, and independent connection retain their positive-node stencils and are not assigned regular-center parity. All positive-radius nodes remain evolved. With that closure, all three puncture grids $N = 257, 513, 1025$ reach $t = 0.2 = 0.6M$. The horizon-exterior Hamiltonian RMS values are 1.623×10^{-3}, 2.767×10^{-4}, and 1.299×10^{-5}; the interior maxima are 7.508×10^{-4}, 5.665×10^{-5}, and 2.706×10^{-6}. Full-domain mixed-profile self-differences give an approximate order of 3.69 at this short common time. The lapse floor does not activate in these resolved-grid tests. This does not assess the very small radii sampled by the earlier stretch-16 grid, nor establish convergence through $100M$.

Verified continuations reach $t = 3 = 9M$ on both $N = 257, 513$. At this common time, exterior Hamiltonian RMS decreases from 6.822×10^{-3} to 5.865×10^{-4} and momentum RMS from 1.362×10^{-2} to 1.154×10^{-3}. The fine-grid final relative horizon-mass error is -8.66×10^{-10}, with maximum sampled drift 4.21×10^{-8}. No third grid has yet reached this duration, so these improvements are not a long-time convergence certificate.

The coarse trajectory continues to $t = 8 = 24M$, beyond the earlier source-control failures. Its final relative horizon-mass error is -2.98×10^{-6}, maximum sampled mass drift 6.42×10^{-6}, and exterior Hamiltonian RMS 5.40×10^{-3}. However, the minimum metric has fallen to 0.459 at the first positive node, and the interior Hamiltonian maximum is 1.12×10^{2} near $r = 0.0156$, inside the horizon at $r \simeq 0.154$. These are unresolved signs of interior deterioration. They are retained as diagnostics rather than used as an interior constraint budget. The physical geometry remains valid at the terminal checkpoint, and no floor activation occurred.

All these runs maintain relative nullness within 2.23×10^{-16}. The sampled quantity $g(c_S^2 - c_L^2)$ remains positive, with minima about 0.260 and 0.267 in the coarse and fine continuations. This excludes a sampled lapse/shift speed crossing along those trajectories; it does not prove uniform eigenvector bounds. At this stage, longer matched refinement was needed to determine whether the late inner degradation moves or decreases with resolution. The $100M$ stability and convergence goal remains open.

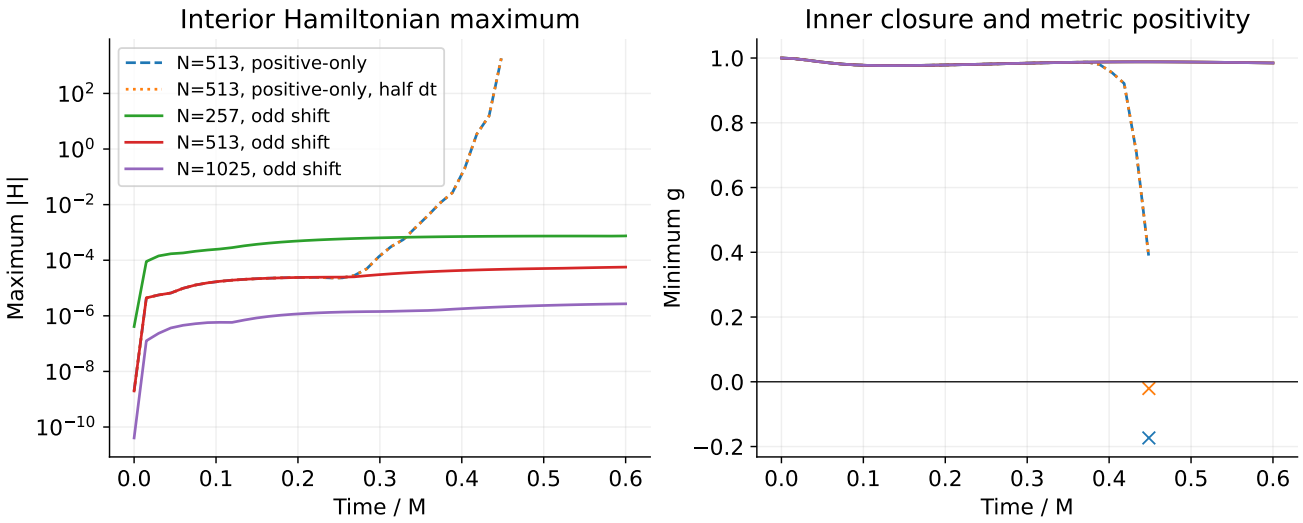


Figure 9: The fast Gamma wave exposes an unstable positive-only inner gauge stencil. Halving the timestep leaves the early failure nearly unchanged. Odd extension of the shift variables removes this short-time failure on three grids; puncture geometry retains its positive-node stencils. Crosses show actual failed-stage metric minima.

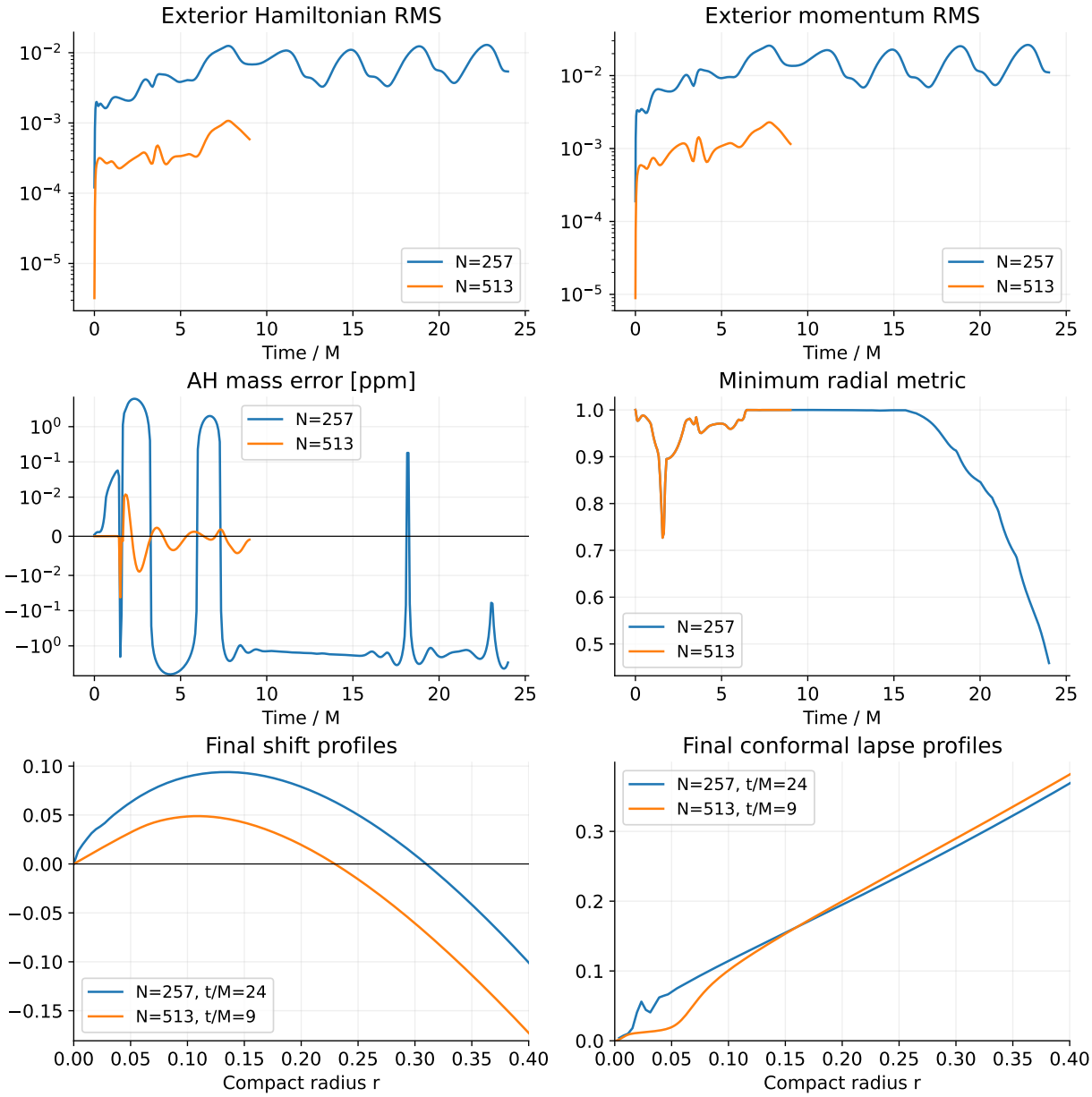


Figure 10: Unsuppressed bulk Gamma-driver continuations with the revised inner gauge closure. Exterior constraints are normalized over apparent-horizon-exterior radial weights. The coarse run survives to $24M$, but its metric minimum deteriorates after about $15M$. Final inner shift and lapse snapshots are at unequal times, as labeled, and are not a convergence comparison. The outward interior shift is now allowed to develop.

6.3 Late inner deterioration and lapse-adjusted Z4 damping

The finest Gamma-driver grid now reaches coordinate $t = 1 = 3M$. At this common time, the exterior Hamiltonian RMS values on $N = 257, 513, 1025$ are 4.1685×10^{-3}, 3.6323×10^{-4}, and 1.4329×10^{-5}. Momentum RMS decreases from 1.0183×10^{-2} to 1.0776×10^{-3} to 5.3229×10^{-5}. Full-domain radial-weighted mixed-profile self-differences are 2.4497×10^{-3} and 1.8103×10^{-4}, giving an approximate order of 3.758. This extends the three-grid evidence to $3M$, not to the duration target.

The unmodified-damping $N = 513$ continuation also reaches $24M$. Its exterior Hamiltonian and momentum RMS values are 3.9834×10^{-4} and 9.8699×10^{-4}, and its maximum sampled horizon-mass drift is 3.13×10^{-7}. At this same time, however, $g_{\min} = 0.504$ and the interior Hamiltonian maximum is 268.1, compared with 0.459 and 111.6 on $N = 257$. The late interior error does not decrease in this maximum norm. Good exterior diagnostics therefore coexist with unresolved inner deterioration on both grids.

To diagnose the later coarse-grid deterioration, we decompose the implemented metric rate as

$$\partial_t g = \beta\, \partial_r^{\mathrm{up}} g - 2\alpha A_{rr} + \frac{4g}{3}(\beta' - \beta/r) + \mathrm{KO}_g. \tag{26}$$

At the first positive node at $24M$, the transport, curvature, shift-shear, and dissipation terms are respectively 0.0335, 0.0430, -0.5064, and -0.0218, giving $\partial_t g = -0.4517$. The metric-derived connection is $V^r = -756.04$, whereas the independent connection is $\Lambda^r = 0.996$. The decomposition agrees with the computed rate within 1.2×10^{-15} in the inspected snapshots. Thus large connection-constraint errors accompany a shift-shear contribution that drives the metric downward. This correlation does not establish a unique nonlinear cause.

A restricted Jacobian on the inner twelve positive nodes at this snapshot, holding the exterior fixed, has maximum eigenvalue real part -0.0516 with the current closure. Adding even metric parity instead gives $+0.4691$; imposing odd connection parity alone gives -0.0516. Two perturbation sizes agree. We therefore do not adopt additional geometry parity. The artificial interface, omitted exterior response, and possible nonnormal growth prevent this restricted calculation from proving full-domain stability.

A separate asymptotic calculation identifies a constraint-growth mechanism in the retained equations. Before flooring changes the AF expansion, write $g = 1 + g_1 r + \cdots$, $\beta = b_1 r + b_2 r^2 + \cdots$, and $\Lambda^r = \Lambda_0 + \cdots$. Then

$$V_0^r = 4g_1, \qquad \dot{g}_1 = b_1 g_1 + \frac{4}{3} b_2, \tag{27}$$

$$\dot{\Lambda}_0 = 2b_1\Lambda_0 - 4b_1 g_1 + \frac{16}{3} b_2, \tag{28}$$

$$\delta = \Lambda_0 - 4g_1, \qquad \dot{\delta} = 2b_1\delta. \tag{29}$$

The native damping vanishes with the collapsed lapse, while outward shift can amplify this leading constraint error. The coarse $24M$ estimate is $b_1 = 3.736$. The expansion motivates a control; it is not a quantitative fit to the finite first node or a late trumpet asymptotic proof.

Lapse-adjusted damping is motivated by the black-hole tests in Sec. III F, Eq. (27), of Alic, Kastaun, and Rezzolla (2013). Our additional compact window and reference-lapse factor adapt that idea to the present hyperboloidal equations. The optional control replaces the damping product by

$$\alpha\kappa_1 \longmapsto (1-w)4\alpha + w\kappa_{\mathrm{in}} A, \qquad \kappa_{\mathrm{in}} = 8, \tag{30}$$

$$d_\kappa = w(\kappa_{\mathrm{in}} A - 4\alpha), \tag{31}$$

$$\Delta S_p = d_\kappa(1-\kappa_2)\Theta, \qquad \Delta S_\Theta = -d_\kappa(2+\kappa_2)\Theta, \tag{32}$$

$$\Delta S_\Lambda = -d_\kappa(\Lambda^r - V^r). \tag{33}$$

Here S is the native pole numerator: the added rates are $\Delta S/\Omega$. The value $\kappa_2 = -0.99$ is unchanged. The implementation evaluates these finite products without dividing by the lapse. Its leading AF constraint equation is $\dot{\delta} = (2b_1 - \kappa_{\rm in})\delta$. The coefficient eight exceeds the measured coarse leading growth estimate $2b_1 \simeq 7.47$; this local criterion does not guarantee nonlinear stability. The correction vanishes identically for $r \geq 0.8$. It changes no principal coefficient or boundary datum, and does not project any constraint. The bulk Gamma driver and telegrapher lapse retain their equations.

An independent native-kernel evaluation with the corresponding spatially varying κ_1 agrees with the added numerator within 2.8×10^{-14}. The AF rate limit, exact equilibrium, independent connection, and stage nullness pass their checks; nine relevant CTest groups pass. At identical states the lapse, relaxation flux, direct shift rate, and entire outer-collar rate are unchanged. New trials start from fresh wormhole data; all earlier trajectories continue with their original frozen damping. The input parameter defaults to zero, which preserves that original damping.

Fresh adjusted-damping puncture runs on $N = 257, 513$ both pass $0.6M$. At that common time, exterior Hamiltonian RMS decreases from 1.170×10^{-3} to 1.947×10^{-4}. The coarse continuation reaches $24M$ with $g_{\rm min} = 0.912$, compared with 0.459 under original damping. Its exterior Hamiltonian RMS is 3.961×10^{-3} and its relative horizon-mass error is -2.177×10^{-6}. However, its interior Hamiltonian maximum is 259.8, larger than the original coarse value 111.6. A positive leading AF damping margin, $\kappa_{\rm in} - 2b_1 = 3.87$, therefore does not prevent finite-radius constraint growth. The outcome is mixed: improved metric positivity at this time, without a demonstrated repair of the inner constraint error.

Both adjusted-damping grids also reach the common time $9M$. Exterior Hamiltonian RMS decreases from 4.791×10^{-3} on $N = 257$ to 3.619×10^{-4} on $N = 513$, and momentum RMS from 9.813×10^{-3} to 7.411×10^{-4}. The fine-grid maximum sampled horizon-mass drift is 4.21×10^{-8}, with final relative error -1.21×10^{-9}. A third grid and an independent puncture timestep study for this modified damping remain outstanding.

The adjusted coarse trajectory subsequently completes $30M$. Its final $g_{\rm min} = 0.885$, exterior Hamiltonian RMS is 5.824×10^{-3}, exterior momentum RMS is 1.161×10^{-2}, and maximum sampled relative horizon-mass drift is 7.08×10^{-6}. Its interior Hamiltonian maximum reaches 537.5. Relative nullness remains within 2.23×10^{-16} and the lapse floor never activates. Interior magnitudes alone are not rejection criteria, but no claim of long-time convergence follows from this single surviving grid.

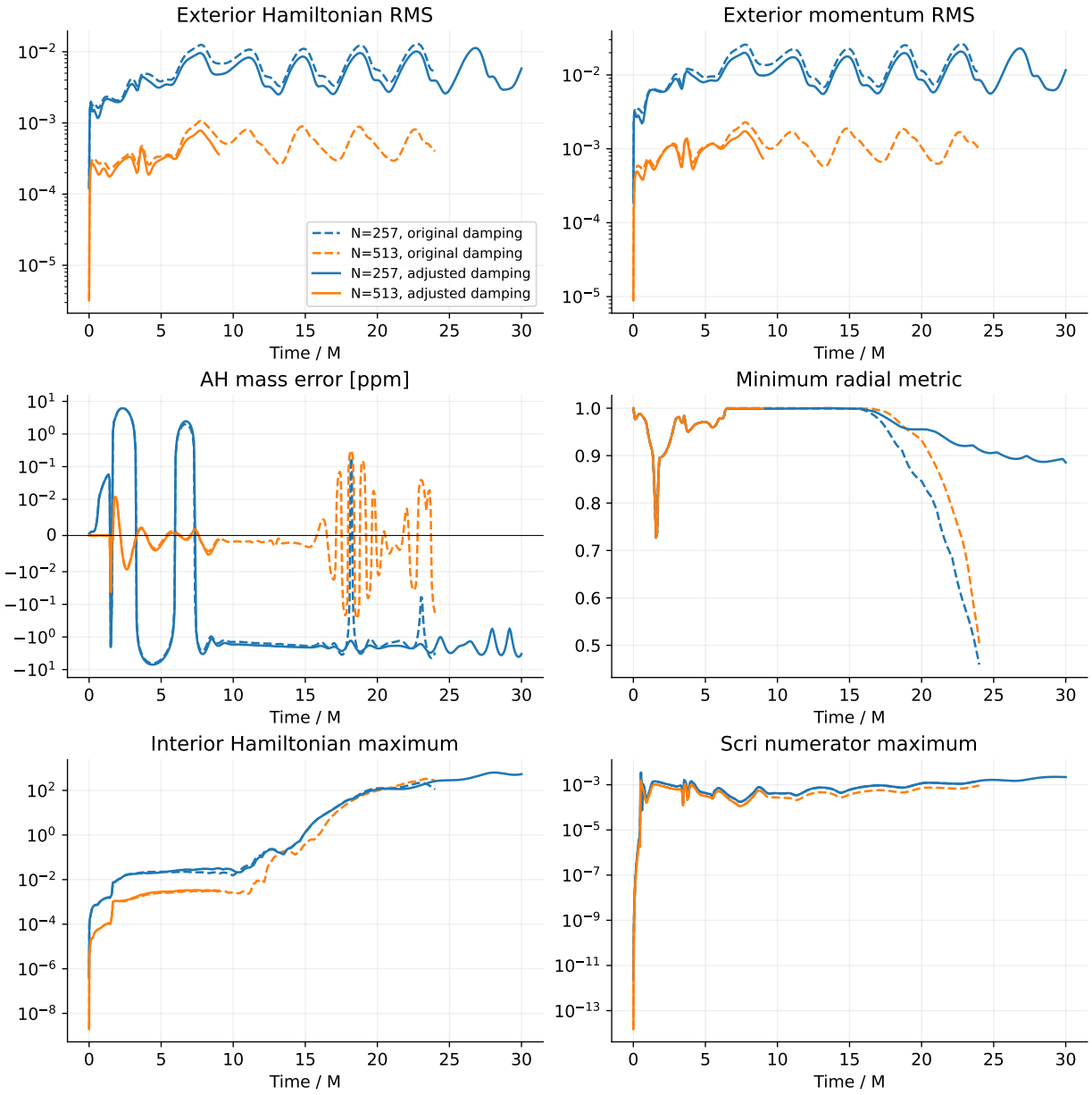


Figure 11: Original and lapse-adjusted Z4 damping with the same bulk Gamma driver. Dashed curves retain original damping; solid curves use the compact adjusted product. Exterior norms use only apparent-horizon-exterior radial weights. The interior maximum is shown separately and is not included in that budget. Improved metric positivity on the adjusted coarse trajectory does not imply reduced interior Hamiltonian error or long-time convergence.

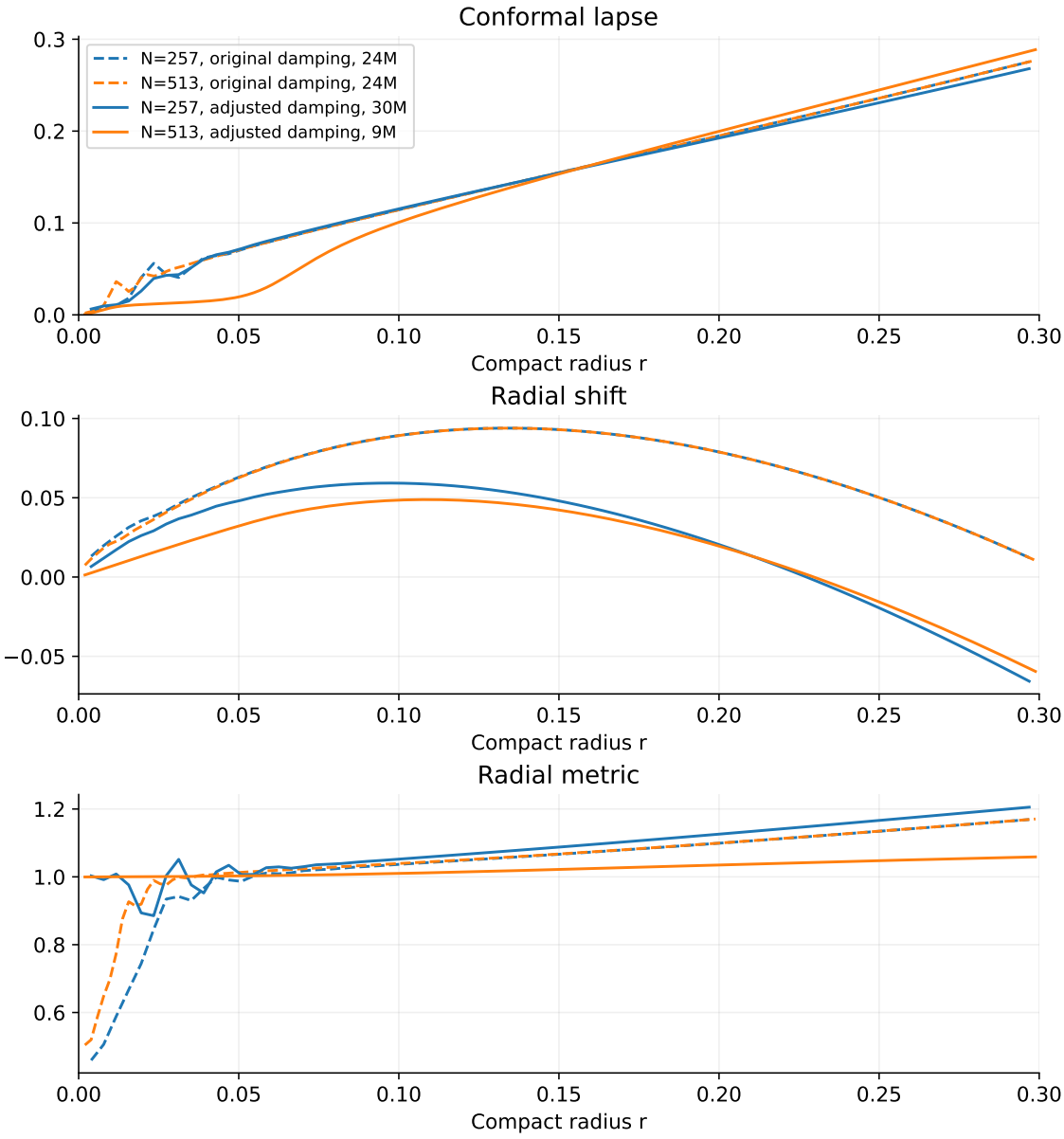


Figure 12: Terminal inner lapse, shift, and metric profiles for the same controls. Snapshot times are stated in the legend; unequal-time profiles are not a convergence test. All positive-radius nodes are retained, and no metric projection or floor has been applied. The configurable physical lapse floor remains 10^{-10} but has not activated in these runs.

6.4 A later exterior speed coincidence and timestep restriction

The coefficient-one adjusted-damping trajectory was continued beyond the preceding $30M$ checkpoint without changing its frozen equations or timestep. It stops at coordinate $t = 15.175 = 45.525M$ on the scri-numerator guard. Its relative nullness still remains within 2.23×10^{-16}. Thus the earlier $30M$ result remains a valid duration milestone, but this longer trial fails qualification.

A distinct defect precedes the terminal failure. Define the scalar gauge-speed margin

$$G(r) = g(c_S^2 - c_L^2) = \frac{4\mu_\Gamma}{3} - \chi\left(F + g\eta\kappa w^2/\tau\right). \tag{34}$$

The first nonpositive sampled minimum occurs at coordinate $t = 10.325 = 30.975M$. At the terminal checkpoint, $t = 15.175$, the minimum is -1.290 near $r = 0.652$. Linear interpolation of adjacent physical states gives speed coincidences near $r = 0.467$ and 0.847, both outside the apparent horizon. At each interpolated root, the positive repeated speed has algebraic multiplicity two but geometric multiplicity one. The next-smallest singular values are 0.0856 and 0.0159, while the smallest are below 2×10^{-16}. This is evidence of a defective scalar symbol on the interpolated trajectory, not a demonstration that crossing characteristic rays caused a lapse shock. It also does not isolate the sole cause of the later nonlinear failure.

An exact minor confirms the interior degeneracy independently of the numerical rank tolerance. On $c_S^2 = c_L^2 = z^2$, delete the p/Ω row and χ_r column from $P - zI$. The resulting minor simplifies to

$$\mathfrak{m} = \frac{2\alpha\chi^2 F z^2}{3g^3}(gz^2 - \alpha^2\chi)^2 > 0 \tag{35}$$

for positive interior data with $c_L^2 > \alpha^2\chi/g$. Meanwhile the determinant vanishes. Hence the rank is nine, and each repeated nonzero gauge speed has geometric multiplicity one. The retained slicing has $F > \alpha^2$ wherever $\Omega, \alpha > 0$, so this condition applies to the diagnosed exterior coincidences. This scalar result does not settle the singular puncture limit or the 3D system.

The next controlled change retains a constant bulk Gamma coefficient and sets $\mu_{\text{bulk}} = 2$. To motivate that value, write $\mathcal{L} = \chi(F + g\eta\kappa w^2/\tau)$ and use $\mu_{\text{separated}} = 3\mathcal{L}/2$. Where $C < 1/2$, positive margin requires

$$\mu_{\text{bulk}} > \frac{3}{4}\frac{1-2C}{1-C}\mathcal{L}. \tag{36}$$

The largest right-hand side on the saved failed trajectory is 1.9675. Two would therefore separate the speeds on those saved states. This counterfactual check does not predict the changed trajectory, which must be tested independently. The bulk remains an advective Gamma driver, and the outer algebraic closure, compact Z4 damping, telegrapher lapse, and physical lapse floor remain unchanged. The exact Minkowski scalar eigenspaces are complete for coefficient two. The previously stated collapsed-corner and 3D limitations still apply.

New puncture trials record the minimum margin over every evaluated RK stage, together with its location. Adding this diagnostic leaves a replayed RHS and RK step bitwise unchanged. It does not project, filter, or alter the evolving fields.

The coefficient-one adjusted-damping family also supplies a separate short puncture timestep study. At $N = 257$ and common coordinate $t = 0.9 = 2.7M$, $\Delta t/h = 0.0375, 0.01875, 0.009375$ gives exterior mixed-profile differences 1.560×10^{-8} and 1.031×10^{-9}, corresponding to order 3.920. The matched two-grid spatial difference at this time is 2.632×10^{-3}, much larger than either temporal difference. The full-domain temporal order is only 1.611, with differences 1.751×10^{-8} and 5.731×10^{-9}; we do not claim fourth-order temporal convergence at the puncture. The three-grid short spatial test at $0.6M$ has order 3.696 with the same damping. Neither short test extends the duration qualification.

The initially attempted $\Delta t/h = 0.075$ run instead loses metric positivity at coordinate $t = 0.0140625$, near $r = 0.99776$. It is retained as a failed control. A restricted linearization of the outer sixteen nodes at the initial state gives a negative eigenvalue about -11604.1. RK4 amplifies that mode by 2.396 at the failed timestep, while the smaller tested timesteps lie inside RK4's stability region for this restricted operator. Thus the characteristic-speed CFL check alone does not ensure stability of the complete boundary discretization. This frozen, restricted spectrum is an explanation consistent with the observed failure, not a full-domain nonlinear stability proof. All duration trials retain $\Delta t/h = 0.025$.

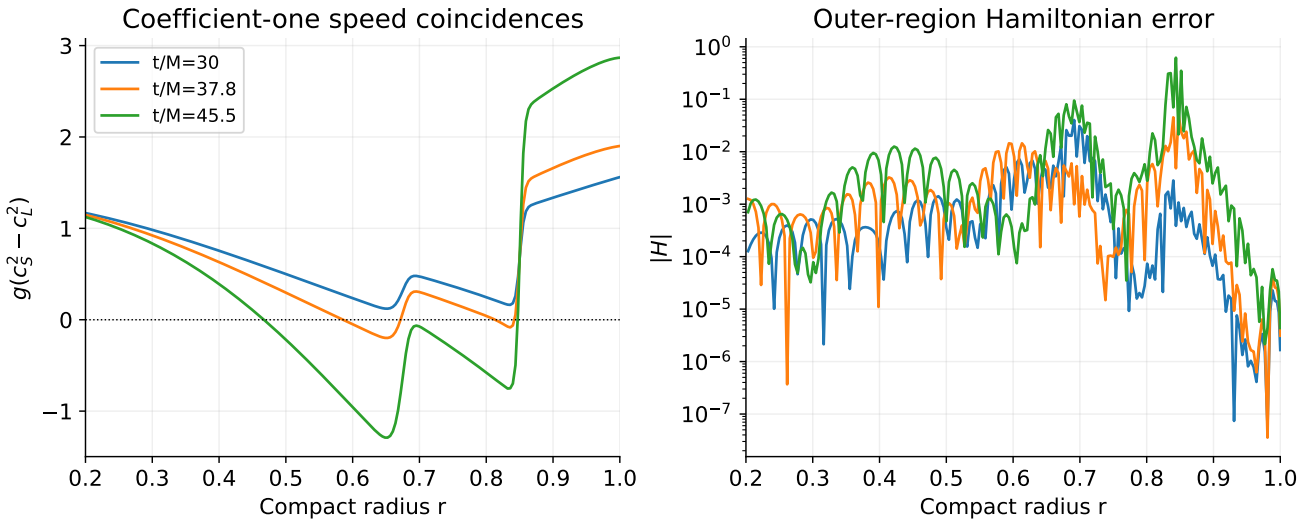


Figure 13: The failed coefficient-one trajectory. The gauge-speed margin becomes negative in the exterior, with two defective speed coincidences at the terminal checkpoint. Hamiltonian error is shown only for $r \geq 0.2$ to display the outer-region structure. Coexistence of the symbol defect and increasing error does not establish a unique nonlinear cause.

6.5 Coefficient-two refinement and later failure

The new coefficient-two family passes the regular-center gate and the three puncture grids through $0.6M$. All three grids subsequently reach $3M$. At nine exact saved times between $0.6M$ and $3M$, full-domain mixed-profile spatial orders range from 3.862 to 4.660; the corresponding exterior orders are nearly identical. At $3M$, the exterior Hamiltonian RMS values on $N = 257, 513, 1025$ are 4.189×10^{-3}, 2.565×10^{-4}, and 6.487×10^{-6}. The exterior mixed-profile differences are 1.055×10^{-3} and 4.456×10^{-5}, giving an instantaneous order of 4.565. These are measured orders over a short trajectory, not a universal formal order or a long-time convergence certificate.

An independent coefficient-two puncture timestep study at $N = 257$, $t = 0.9$, and $\Delta t/h = 0.0375$, $0.01875, 0.009375$ gives exterior mixed-profile differences 2.192×10^{-9} and 1.362×10^{-10}, corresponding to order 4.009. Full-domain differences are 2.196×10^{-9} and 1.423×10^{-10}, with order 3.948. These temporal differences are much smaller than the measured spatial differences on the same short trajectory. Independent temporal accuracy at the eventual duration target remains to be established.

The $N = 513$ coefficient-two trajectory reaches $9M$, with exterior Hamiltonian RMS 3.726×10^{-4} and maximum sampled relative horizon-mass drift 2.42×10^{-9}. At the same time the coarse Hamiltonian RMS is 3.985×10^{-3}. No third grid has yet reached this duration.

The coarse trajectory completes $72M$ but fails the subsequent extension at $t = 24.9 = 74.7M$ on the scri-numerator guard. At the stop, exterior Hamiltonian RMS is 0.3206, the numerator is 0.01153, and

$g_{\min} = 0.1455$ remains positive. Maximum sampled horizon-mass drift is 1.08×10^{-4}, showing that mass conservation alone would not expose this failure. The first report retaining a nonpositive stage speed margin is at $72.225M$; the recorded minimum is eventually -0.0855. The final instantaneous margin has returned to $+0.499$, so inspecting only the terminal profile would miss the transient coincidence. The coefficient-two control therefore delays, but does not remove, this qualification failure.

Future trials stop on a nonpositive recorded speed margin and preserve the first affected finite state with its RHS-call index, location, and hash. An integration check rejects a deliberately defective initial state before advancing time. Offline replay also captures the actual coefficient-two counterexample at coordinate $t = 24.1$. These are stop conditions, not field corrections; no failed trajectory is restarted.

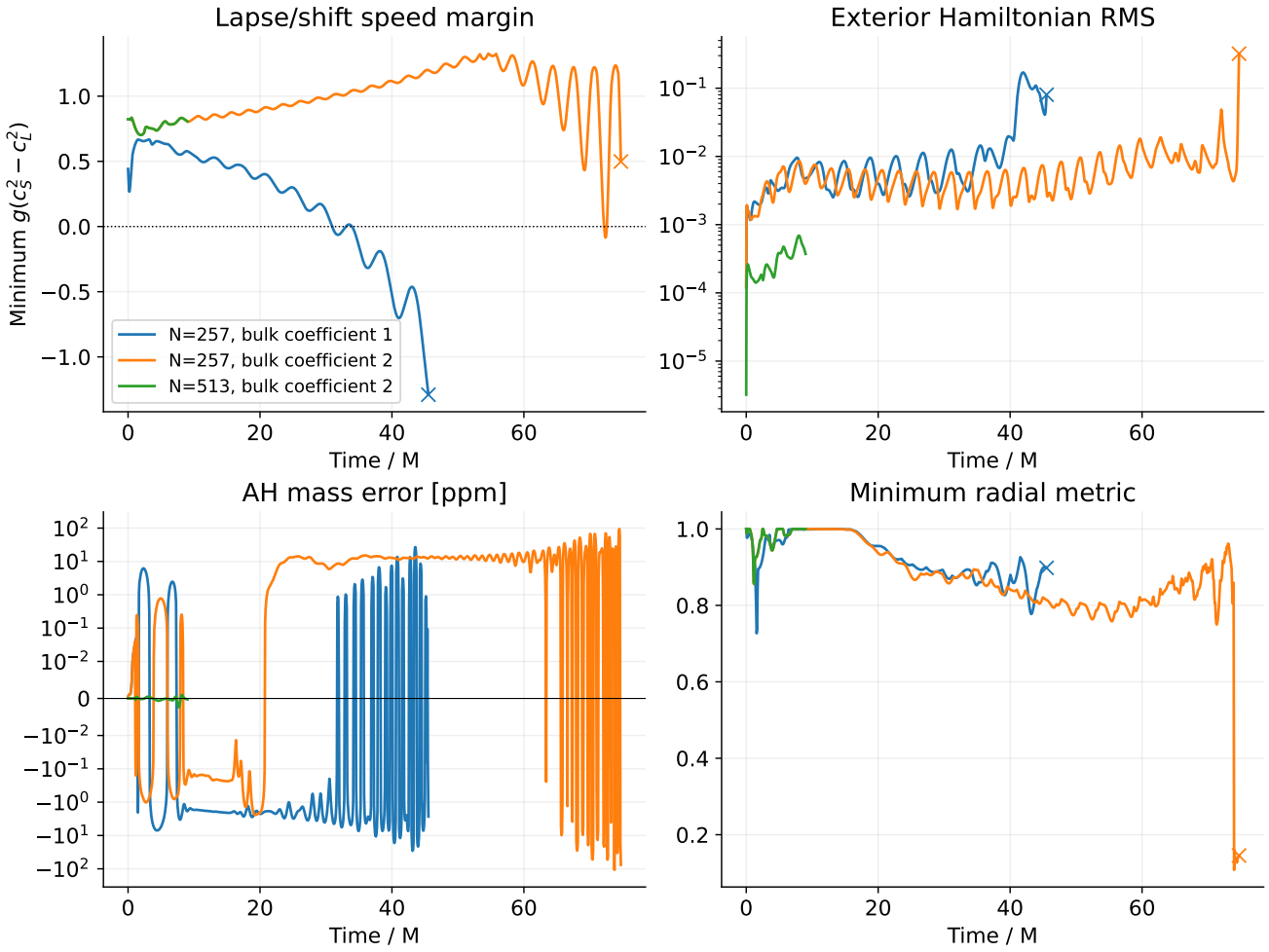


Figure 14: Constant-coefficient controls with the same compact Z4 damping. Crosses mark failed trajectories' last reported states. The plotted speed margin is instantaneous; the coefficient-two run's final positive value does not erase its earlier negative stage margin. Both coarse runs fail before $100M$, while the finer coefficient-two run shown here reaches $9M$.

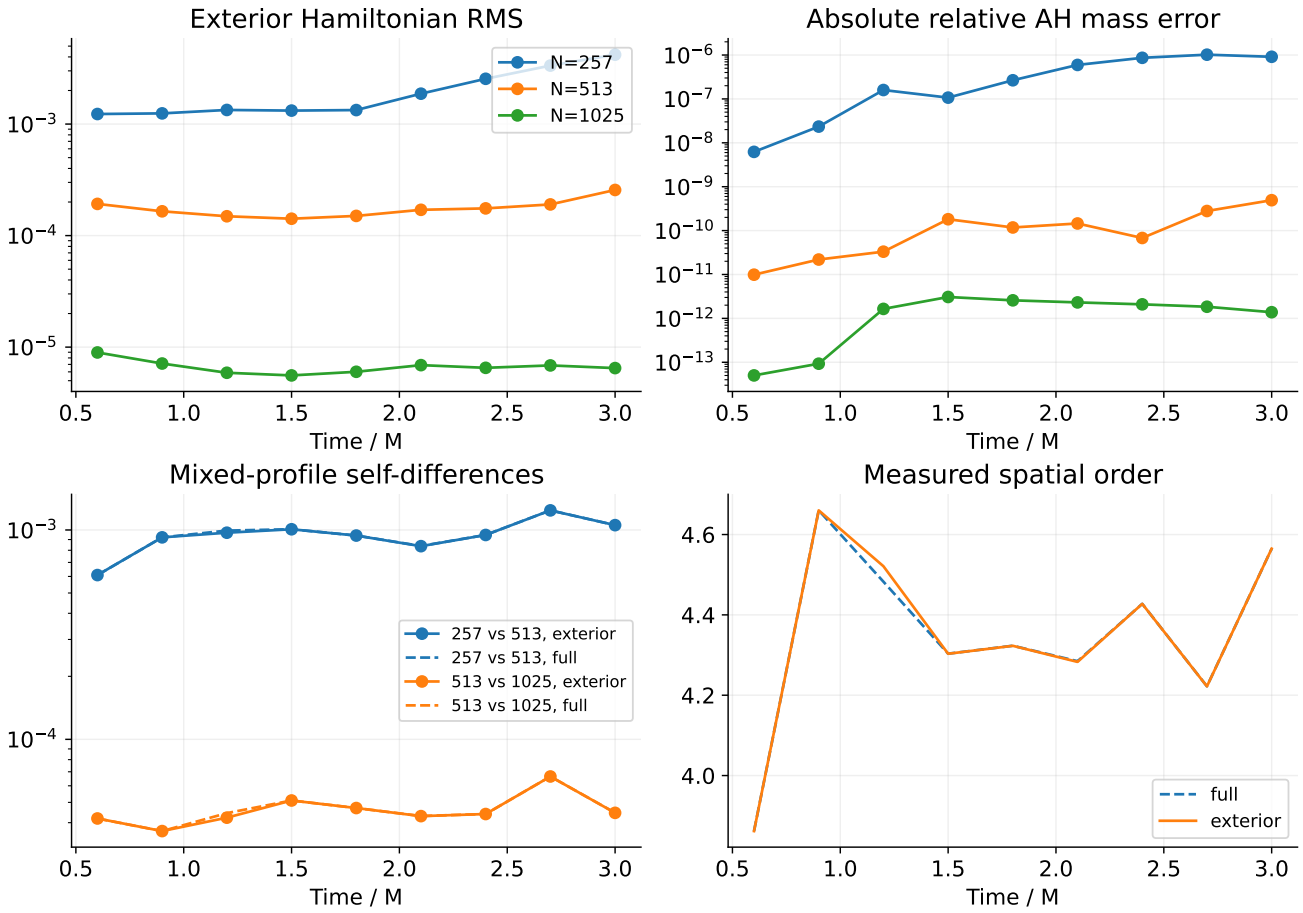


Figure 15: Coefficient-two three-grid refinement at nine exact common times. Full and exterior mixed-field differences are nearly coincident on this short trajectory. The exterior mask for each pair begins at the larger of its two interpolated horizon radii; radial weights are normalized over the retained region. The field set is $(\chi, g, A_{rr}, p+3, \Theta, \Lambda^r, \alpha, \beta, B)$.

6.6 A fresh coefficient-four control

The coefficient-two failure motivates a fresh trial with constant $\mu_{\mathrm{bulk}} = 4$. The largest coefficient required to keep the margin positive on its saved final segment was 2.05482. Four supplies a larger counterfactual margin, while retaining the advective bulk Gamma-driver equations; it is not a guarantee about the new trajectory. The exact Minkowski scalar eigenspaces remain complete. The compact damping, telegrapher lapse, physical lapse floor, outer collar, and algebraic boundary traces are unchanged.

All four regular-center controls complete their prescribed coordinate interval $t = 1$. The three fresh puncture grids reach $t = 0.2 = 0.6M$. Their exterior Hamiltonian RMS values are 1.251×10^{-3}, 1.918×10^{-4}, and 8.898×10^{-6} on $N = 257, 513, 1025$. Exterior mixed-profile differences are 4.690×10^{-4} and 2.983×10^{-5}, giving order 3.975 at this common time; the full-domain order agrees to five decimal places. No lapse-floor activation occurs. These results qualify a short test of coefficient four only; the coefficient-two temporal results cannot be transferred to this changed system.

All three coefficient-four grids subsequently reach $9M$. At 29 exact common times from $0.6M$ to $9M$, exterior mixed-profile orders range from 3.900 to 4.808; full-domain orders range from 3.891 to 4.797. Exterior Hamiltonian and momentum norms, and absolute relative horizon-mass errors, decrease with resolution at every sampled time. At $9M$, the Hamiltonian RMS values are 4.341×10^{-3}, 3.767×10^{-4}, and 7.072×10^{-6}. The exterior profile order is 4.653, compared with full-domain order 4.504. Peak sampled relative horizon-mass drifts over the common interval are 1.004×10^{-6}, 2.813×10^{-9}, and 3.559×10^{-11}. The collector verifies fixed equations, resolution, and timestep along each trajectory, and requires bitwise agreement where saved continuation segments overlap.

Subsequent matched profiles extend the spatial comparison to $12.6M$. At seven exact times from $10.8M$ to $12.6M$, exterior Hamiltonian and momentum norms and absolute relative horizon-mass errors decrease with resolution at every sample. The historical nine-field exterior orders range from 4.170 to 4.750, but the full-domain order falls to 1.979 at the last sample. At $12.6M$, exterior Hamiltonian RMS values are 3.610×10^{-3}, 1.379×10^{-4}, and 2.960×10^{-6}; momentum RMS values are 7.318×10^{-3}, 3.049×10^{-4}, and 9.334×10^{-6}. The corresponding absolute relative horizon-mass errors are 5.495×10^{-8}, 3.572×10^{-11}, and 1.215×10^{-12}.

A supplemental regional audit includes the auxiliary shift driver D as a tenth field and reproduces the historical nine-field norms before extending them. With each regional norm normalized over its own radial weights, the ten-field exterior orders over this interval are 3.743–4.627. At $12.6M$, the full-domain order is 2.031, the horizon-interior order is 1.907, and the puncture-region order for $r \leq 0.02$ is 1.357. The largest interior self-differences are in A_{rr} and $p + 3$; this localizes the reduced full-domain order without establishing its cause. The scri collar $r \geq 0.9$ gives orders 1.445–1.508, with a substantial contribution from D. Exact nullness therefore coexists with lower-order convergence of other boundary fields. These local results remain visible even though the primary physical budget excludes the horizon interior. No nodes are removed from the full-domain assessment, and no uniform fourth-order claim is made.

Independent $N = 257$ controls with $\Delta t/h = 0.05, 0.025, 0.0125$ all reach $9M$ from the same initial data. At $3M, 6M, 9M$, exterior mixed-profile temporal orders are 3.983, 3.918, and 3.946. At $9M$, the successive temporal differences are 2.465×10^{-11} and 1.599×10^{-12}, far below the same-time spatial difference 3.012×10^{-4} between $N = 257$ and 513. The full-domain temporal order at $9M$ is only 3.407, with differences 4.634×10^{-11} and 4.370×10^{-12}; we do not promote the exterior order to a uniform full-domain result. Independent temporal accuracy and matched spatial refinement through $100M$ remain outstanding.

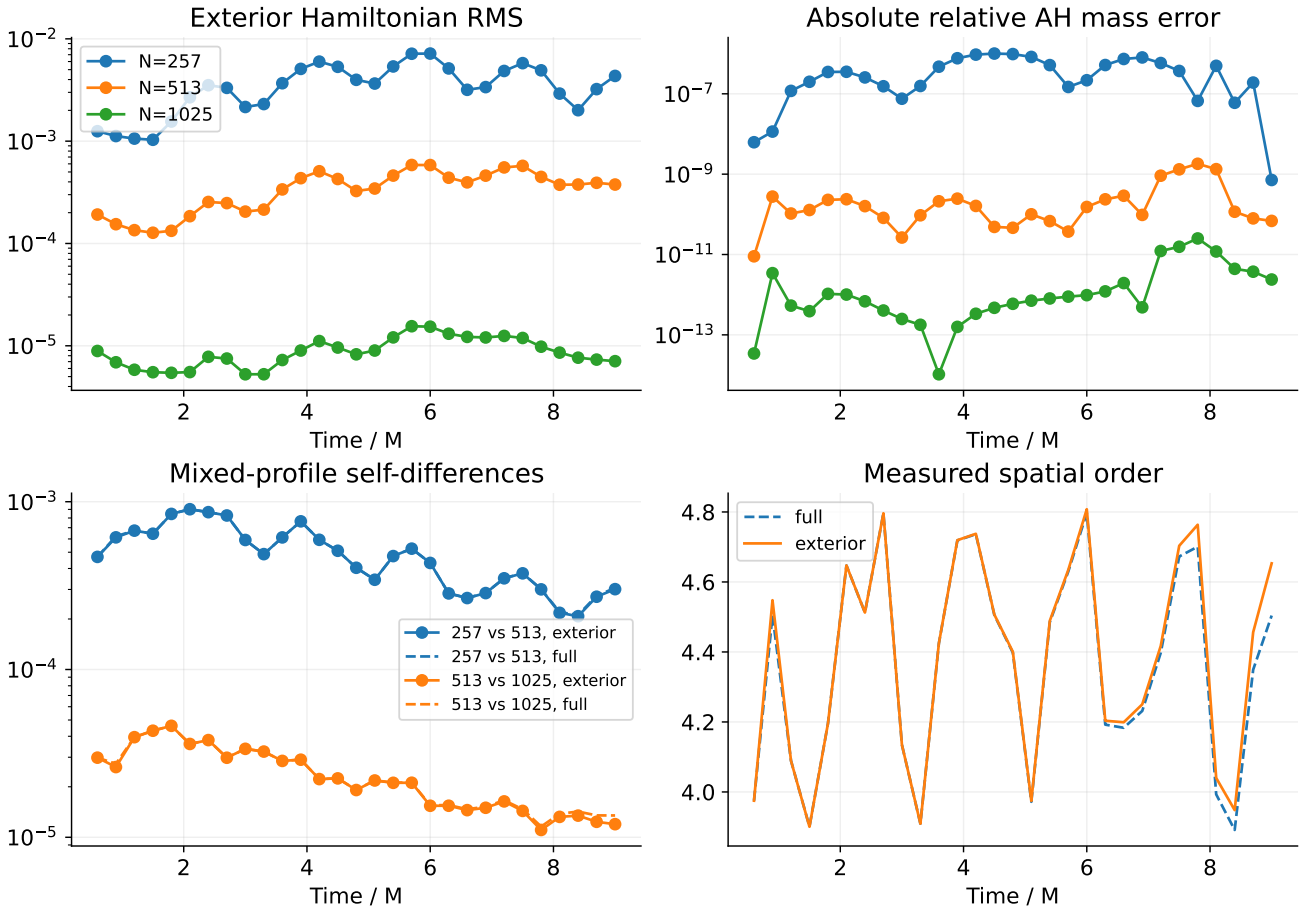


Figure 16: Coefficient-four spatial refinement through $9M$ at 29 exact common times. Both exterior constraints and horizon-mass errors decrease with resolution. Full and exterior profile differences are distinguished; their measured orders are not assumed to hold beyond this interval.

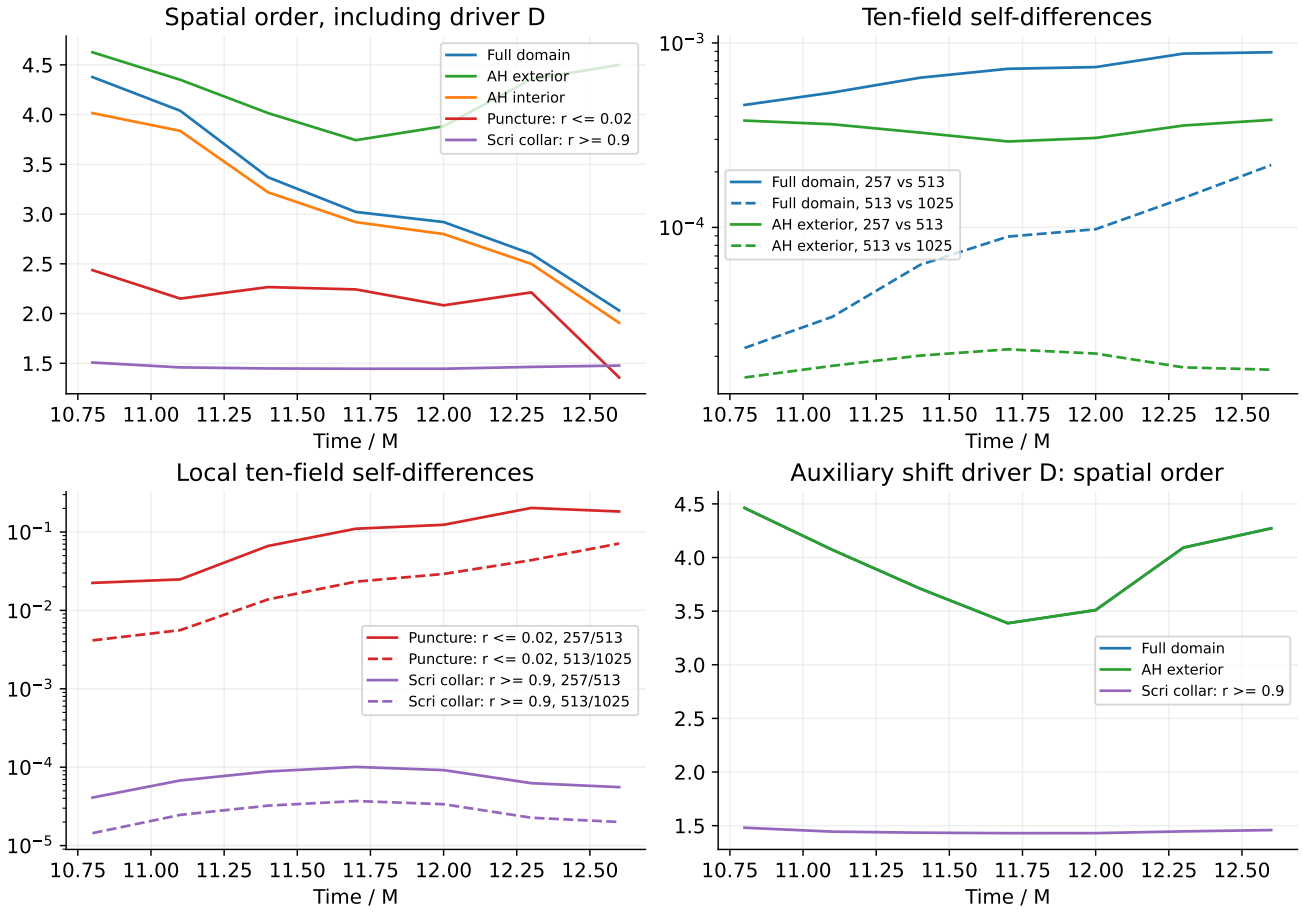


Figure 17: Regional spatial refinement at seven exact common times from $10.8M$ to $12.6M$, including the auxiliary shift driver D. The full-domain order declines as interior curvature differences grow. The scri collar also has lower order than the aggregate exterior norm. Solid and dashed difference curves compare 257/513 and 513/1025 nodes, respectively; all norms are normalized within their labeled regions. These are finite-resolution self-convergence measurements, not asymptotic order proofs.

The coarse coefficient-four trajectory completes coordinate $t = 33.3333984375$, or $100.0002M$, without a guard failure. Its final exterior Hamiltonian and momentum RMS values are 2.344×10^{-3} and 4.462×10^{-3}. The radial metric minimum is 0.7518, and the minimum speed margin over every evaluated stage remains 0.7189. Maximum sampled relative horizon-mass drift is 1.655×10^{-4}; the final relative mass error is -2.563×10^{-5}. Relative scri nullness remains within 2.23×10^{-16}, and the lapse floor never activates. This establishes a coarse duration result for the frozen equations and timestep, not three-grid stability or convergence through $100M$. The interior Hamiltonian maximum is still 8.160×10^{3} at the end. It remains a secondary diagnostic rather than an exterior rejection criterion; no positive-radius node has been removed.

A read-only extraction check varies the local horizon interpolation width through $3, 5, 7, 9, 11$ nodes using the same frozen states. At $30M$, the relative spread among widths $7, 9, 11$ is 2.51×10^{-7}, compared with a relative mass error of -1.39×10^{-4}. At $100M$, that spread is 1.72×10^{-10}, far below the measured mass error. Interpolation order therefore does not explain the late mass offset by itself. Additional sign changes of the outgoing expansion occur inside the outermost horizon; these are retained as interior diagnostics, and the exterior budget continues to use the outermost horizon. Agreement among interpolation widths does not establish evolution accuracy.

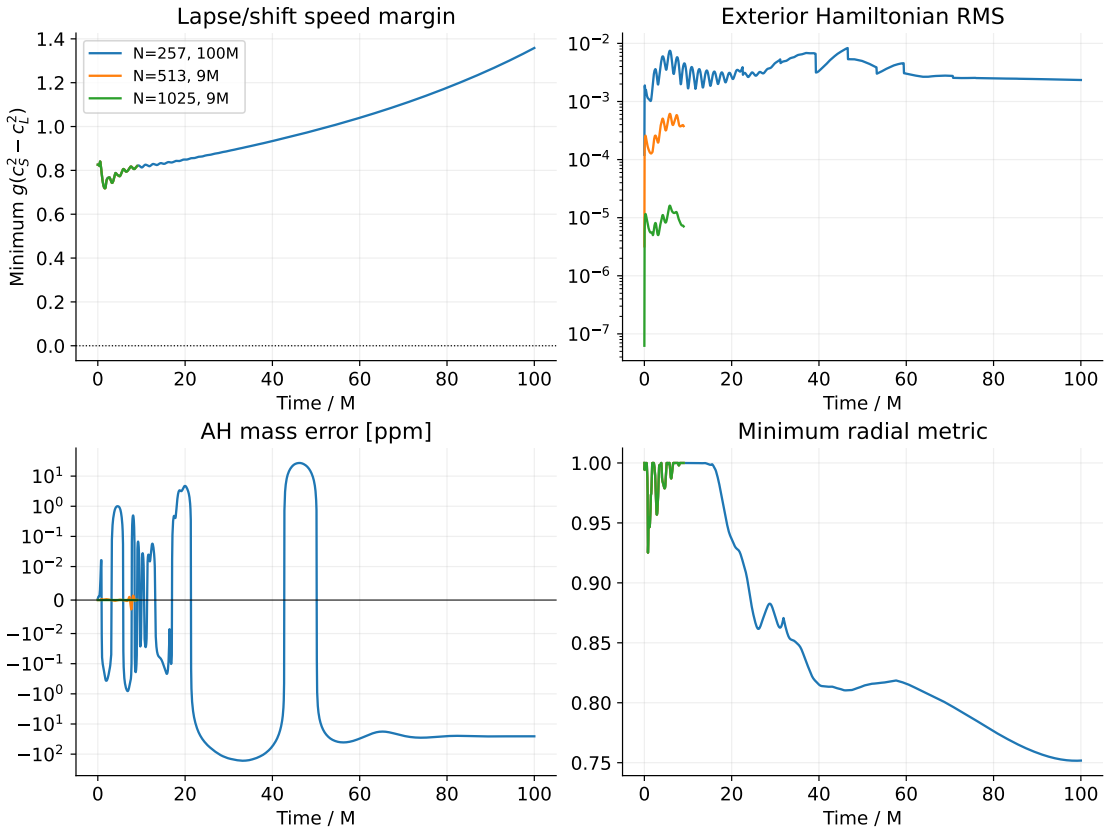


Figure 18: The fresh coefficient-four control. The coarse trajectory completes $100M$; the finer curves in this archived duration snapshot end at their completed $9M$ checkpoints. Subsequent matched spatial evidence reaches $12.6M$, as discussed above. All evaluated stage speed margins remain positive, and the lapse floor has not activated. Unequal-duration histories do not demonstrate long-time convergence; the larger coarse horizon-mass drift remains to be tested at matched finer resolution.

7 Implementation route and reproducibility

A genuine 3D implementation should evolve Cartesian shift and auxiliary vectors in the interior and transform them consistently to shell frames. The spherical code evolves the full radial component, not a globally slaved β/r profile. Spherical symmetry necessarily fixes the central location; it is not a test of puncture motion. Constant and linear Cartesian shift modes must be included in the 3D driver tests. The reference source and restoring term can influence translations, so their effect on puncture tracking must be measured rather than assumed benign.

At a cubed-sphere outer face, compute s_i from the live regular metric and the analytic Ω_i. Eliminate the normal shift component after every stage, ghost fill, prolongation, restriction, and synchronization. Derive the corresponding auxiliary compatibility trace in that frame. Specify tangential coordinates only after counting the vector-sector characteristics. A normal nullness identity alone supplies no tangential gauge condition.

Before deployment, derive a direction-independent 3D version of the telegrapher coupling and speed coefficient, and audit all scalar, vector, and tensor eigenvectors for arbitrary positive metrics. The present radial g-dependent speed separation is not yet such a 3D prescription. AMR needs stage-consistent characteristic interface treatment, stable dissipation across refinement boundaries, and nullness reconstruction after interpolation; a boundary component should not be interpolated as an independent scalar. Shell interfaces additionally require tensor/connection transformation and reference-connection checks. Moving punctures require positive-variable interpolation, measured floor effects, and resolution studies of the near-puncture constraint and gauge fields.

The implementation is under `tst/hyperbolic_shift_1d/`; frozen inputs are under `inputs/hyperbolic_shift_1d/`. Each run records its command, source hashes, library hash, input values, stop reason, histories, and profiles. Failed runs retain their actual outcomes. Portable exports are under `docs/hyperbolic-shift-1d/evidence/`; complete per-run source snapshots and libraries remain in the ignored local output tree. The original elliptic implementation is unchanged except for an additive native adapter with unchanged legacy entry points; all 24 independent $g = 1/\text{tensor}$ comparison cases still pass.

Evidence category	Established scope
Analytic identities	Nullness elimination, evolving-normal rate, scalar principal polynomial, selected exact eigenspaces, leading AF limits.
Algebraic numerics	Every tested stage/state satisfies spherical nullness within one machine epsilon; general 3D boundary identities have separate randomized tests.
Convergence evidence	Short regular-center refinement; coefficient-two three-grid puncture evidence through $3M$; coefficient-four spatial evidence through $12.6M$ and temporal evidence through $9M$, with lower regional orders near the puncture and scri; coarse coefficient-four survival through $100M$, without a $100M$ convergence certificate.
Unresolved	Complete boundary well-posedness, uniform collapsed-lapse hyperbolicity, accurate puncture evolution, long-time robustness, and all 3D/AMR qualification.

The practical outcome is a reproducible experimental alternative with a working exact-nullness mechanism. It is not a replacement for the successful elliptic configuration.